Stochastic Smooth Discrete Local Volatility

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Abstract

We present a statically and dynamically arbitrage-free European option market model with smooth interpolation between call prices, based on the SANOS [6] framework.

In spirit, it is a stocastic model for local volatility when the latter is used as a representation of a European option price surface, not as a pricing model.

We illustrate the strength of our approach on S&P Index Option data from January 2022 to August 2025. We show that the model's initial state fits with in bid/ask for 95% of relevant market option prices.[1] We then train the model dynamics under $\mathbb{P}$ and $\mathbb{Q}$ and show that the model is indeed free of arbitrage while fitting the dynamics of the market well.

1 Introduction

This article presents an arbitrage-free market model for for smooth surfaces of European option prices. While not strictly the first of is kind, it is the first model with an industy-grade ability to fit the current market, allow for smooth interpolation of the entire option surface at each time, and provides an explicit construction of the continuous time risk-neutral drift for both (discrete local) volatilities and spot. It also has a natural estimation strategy for its parameter space.

The core architecture follows the SANOS approach [6] which models the stock price as the product of a time-and-state discrete process X and a smoothing process Y, in our case Black & Scholes. The former is used to encode the option market using a discrete form of local volatilities as code book. That does not mean that the model is a local volatility model – we merely use the implied local volatilities as a convenient representation of the option surface. Compared to using implied volatilities the advantage is that mere positivity of the local volatilities is sufficient to ensure the the option market is free of static arbitrage at all times. We have shown in [6] that this representation can be fitted to real life market data very efficiently using linear programming.

The key contribution in this paper is showing how to construct a dynamic model of SANOS option call prices which are not only statically but also dynamically free of arbitrage. We show the theoretical construction of the corresponding risk-neutral measure $\mathbb{Q}$ and discuss existence of the respective martingale processes. We then discuss implementation and estimate our model on S&P Index Option data from January 2022 to August 2025. We fit our daily model parameters to observed bid/ask prices and then proceed to estimate both $\mathbb{Q}$ and its drift under $\mathbb{P}$. We show

[1] The model presented was trained for daily expiries for 25 business days and strikes between $[\exp(-3\sigma_{atm}\sqrt{t}), \exp(+2\sigma_{atm}\sqrt{t})]$.

that on this data set more than 90% of calibrated model prices fit within observed market bid/ask spreads even with the simplistic grid architecture we present here.

Our approach is closely related to the seminal but unpublished work [21]. In [10] the authors derive related conditions for continuous time local volatility models, but without a tangible implementation strategy. Our results can be seen as a joint extension of the two papers, plus the smoothing from our SANOS [6].

Related Work

Our work is further related to [11] where the authors solve the problem of finding a dynamically arbitrage-free evolution of the full option surface via products of indepdent Levy processes. While structurally very elegant option prices generated by products of independent increment processes do not typically fit well to equity index European option surfaces. Our approach also has the advantage of being numerically accessible. In particular, our model prices can be fit very efficiently within nearly all[2] of observed market bid/ask prices using the linear programming SANOS [6] calibration method.

The model presented here is distinct from our own DYSANOS [7]. There, we focused on learning realistic market dynamics for smooth statically arbitrage-free SANOS surfaces under $\mathbb{P}$. We did also construct an equivalent risk-neutral measure $\mathbb{Q}$ numerically using the *near-martingale measure* approach introduced in [8]. While this method is model-free and can in theory be applied to any option market model under $\mathbb{P}$, it is numerically heavy and in practise restricted to making only a fairly limited number of the overall tradable instruments dynamically arbitrage-free.[3] In the current work we cover all options with an explicit construction of the risk-neutral drift, which is far more efficient than the numerical near-martingale measure approach.

Hans comments

Maybe it's worth to state what we each think where we are and what the gaps are. Please comment–:

1. **DLV risk neutral diffusion and future densitioes** given a front density volatility: I think we are all clear in principle. I also noted you managed to prove existence which is really good. One quesition is what happens if $q \downarrow 0$... do we have an answer to that?

2. **DLV sampling spot** I suppose I was thinking we can sample from the front; that does not mean full step. See below. I think this way front density diffusion and spot are exactly in line. It does mean spot end ups at grid points; I suspect we can't get around that without a guanrateed arb-free call price interpolator.

3. **DLV sampling strategy** to preserve martingale property – testing that.

4. **DLV Increasing grid** I think works but I need to write it down properly ... **Magnus** ?

5. $\mathbb{P}$ **estimation** I have some ChatGPT code but nothing uber-smart. The main thing is that I think we can actually estime at the DLV vol-of-vols indepdenently of spot. Since we know its volatility ξ we know that the ¶ drift needs to be in $(\xi'\xi)^{-1}\xi$ times some vector of appropriate dimension.

[2] In below experiments, above 90%.
[3] In [7] we used 49 instruments per simulated date.

6. **DLV Smoothing** I am in the cheap camp: if X (the DLV process with all its option prices) is a margingale, then XY for Y an independent martingale is also martingale. If Y is (extended) Black & Scholes then I think it needs to be the same under ¶ and $\mathbb{Q}$. In general we can simply use a $\mathbb{Q}$ martingale and estimate the ¶ drift through X alone. Note that products with independent martingales create more of a "U" shape. I think **Blanka** you said you found a way to correlate X with Y?

Assume fixed expiry dates are $0 = T_0 < T_1 < \cdots < T_M$. Assume K are the strikes and that q_t^j is the density for T_j at time $t \le T_j$. I think the core arguments are as follows:

Future Densities

Nothing new in this section

Let T_k be the front expiry, so $T_{k-1} \le t < T_k$, and suppose that q_t^k is known. For $j = k+1, \ldots, M$, let $\Sigma_t^j \in \mathbb{R}^N$ be the vector of discrete local variances into T_j, extend it by zero at the two boundary nodes, and set

$$S_t^j := \operatorname{diag}(0, \Sigma_t^{j,1}, \ldots, \Sigma_t^{j,N}, 0).$$

The future densities satisfy

$$q_t^{j-1} = M_t^j q_t^j, \qquad M_t^j := I - \Omega_j S_t^j, \tag{1}$$

where $\Omega_j \in \mathbb{R}^{(N+2)\times(N+2)}$ is the fixed strike-grid operator for the interval $T_j - T_{j-1}$. The matrix M_t^j is an invertible tridiagonal M-matrix for every non-negative S_t^j.

Let W be a d-dimensional Brownian motion, and designate W^1 as the spot factor: spot is always driven by dW_t^1. Each density coordinate is a martingale. We specify the front-density diffusion as

$$dq_t^k = E_t^{k;1}\, dW_t^1, \qquad E_t^{k;1} := \operatorname{diag}(q_t^k)\eta_t^{k;1}, \qquad E_t^{k;f} := 0 \quad (f = 2, \ldots, d),$$

where $E_t^{k;1} \in \mathbb{R}^{N+2}$ must satisfy $\mathbf{1}^\top E_t^{k;1} = 0$, so that q_t^k retains unit mass. The remaining Brownian factors do not directly move the front density. They drive the stochastic DLV surface, while $\xi_t^{j,\cdot,1}$ determines its leverage with spot. We do not specify η_t^j for $j > k$; the future density volatilities will be obtained recursively.

For future expiries T_j with $j > k$, assume the discrete local volatity squared is given as

$$\frac{d\Sigma_t^j}{\Sigma_t^j} = \mu_t^j\, dt + \xi_t^j\, dW_t,$$

where $\xi_t^j \in \mathbb{R}^{N\times d}$ is given and $\mu_t^j \in \mathbb{R}^N$ is the unknown risk-neutral drift. We write this as

$$dS_t^j = A_t^j\, dt + \sum_{f=1}^{d} B_t^{j;f}\, dW_t^f,$$

with $A_t^j := \operatorname{diag}(0, \Sigma_t^{j,1}\mu_t^{j,1}, \ldots, \Sigma_t^{j,N}\mu_t^{j,N}, 0)$ and $B_t^{j;f} := \operatorname{diag}(0, \Sigma_t^{j,1}\xi_t^{j,1;f}, \ldots, \Sigma_t^{j,N}\xi_t^{j,N;f}, 0)$. With the above

$$dM_t^j = -\Omega_j\, dS_t^j = -\Omega_j A_t^j\, dt - \sum_{f=1}^{d} \Omega_j B_t^{j;f}\, dW_t^f.$$

Setting $dq_t^j = \sum_f E_t^{j;f}\, dW_t^f$ for an unknown $E_t^{j;f}$ yields

$$dq_t^{j-1} = M_t^j\, dq_t^j + (dM_t^j) q_t^j + d[M^j, q^j]_t \tag{2}$$

$$= \sum_{f=1}^{d} \left(M_t^j E_t^{j;f} - \Omega_j B_t^{j;f} q_t^j \right) dW_t^f - \Omega_j \left(A_t^j q_t^j + \sum_{f=1}^{d} B_t^{j;f} E_t^{j;f} \right) dt\ . \tag{3}$$

Matching the diffusion coefficients gives

$$E_t^{j;f} = (M_t^j)^{-1} \left(E_t^{j-1;f} + \Omega_j B_t^{j;f} q_t^j \right) \in \mathbb{R}^{N+2} \quad \text{and} \quad \eta_t^{j,i;f} = \frac{E_t^{j,i;f}}{q_t^{j,i}}\ . \tag{4}$$

The drift must vanish, hence

$$\Omega_j \left(A_t^j q_t^j + \sum_{f=1}^{d} B_t^{j;f} E_t^{j;f} \right) = 0.$$

The restriction of Ω_j to vectors supported on the interior nodes is injective. Therefore the bracket vanishes node by node, giving

$$\mu_t^{j,i} = -\sum_{f=1}^{d} \xi_t^{j,i;f} \frac{E_t^{j,i;f}}{q_t^{j,i}} = -\sum_{f=1}^{d} \xi_t^{j,i;f} \eta_t^{j,i;f}, \qquad i = 1, \ldots, N. \tag{5}$$

Thus the prescribed front-density volatility and the prescribed DLV volatilities determine all future density volatilities and the risk-neutral DLV drifts by forward induction over the expiries.

Front density

The remaining task is to specify both spot dynamics consistent with the front density and the respective front density volatilities.

Case $t = T_k - dt$

Assume we are at $t = T_k - dt$ and observe the front density q_t^k. Current spot is given as $S_t = K^\top q_t^k$. In order to sample a spot consistent with the discrete set of strikes, we can either draw a discrete sample or fit some other distribution exactly to the observed option prices (fitting to densities or probabilities will not work in general).

The simplest method is to draw spot from this discrete distribution using the spot factor W^1. Let F_t^k denote the conditional CDF which assigns probability $q_t^{k,i}$ to strike K_i, and let Φ denote the standard normal CDF. Then

$$S_{T_k} = (F_t^k)^{-1}(\Phi(Z))\,, \qquad Z := \frac{W_{T_k}^1 - W_t^1}{\sqrt{dt}} \sim N(0,1).$$

Correlation between spot and a future DLV coordinate is specified by that coordinate's loading $\xi_t^{j,i;1}$ on W^1.

Conditional on this definition of terminal spot, the front-density diffusion is exact if the Brownian increment is revealed progressively: define

$$Q_i := \sum_{\ell=0}^{i} q_t^{k,\ell}, \qquad a_i := \Phi^{-1}(Q_i), \qquad a_{-1} := -\infty, \quad a_{N+1} := +\infty,$$

and let ϕ be the standard normal density. For $t \le s < T_k$, set

$$z_i(s) := \frac{\sqrt{dt}\, a_i - (W_s^1 - W_t^1)}{\sqrt{T_k - s}}.$$

The time-s front density is the conditional probability

$$q_s^{k,i} := \mathbb{P}(S_{T_k} = K_i \mid \mathcal{F}_s) = \Phi(z_i(s)) - \Phi(z_{i-1}(s)). \tag{6}$$

It is therefore a martingale. Applying Itô's formula gives the exact diffusion

$$dq_s^{k,i} = E_s^{k,i;1}\, dW_s^1, \qquad E_s^{k,i;1} = \frac{\phi(z_{i-1}(s)) - \phi(z_i(s))}{\sqrt{T_k - s}}, \qquad t \le s < T_k, \tag{7}$$

with zero drift. Since $z_i(t) = a_i$, its initial value is

$$E_t^{k,i;1} = \frac{\phi(a_{i-1}) - \phi(a_i)}{\sqrt{dt}} \quad \text{hence} \quad \eta_t^{k,i;1} = \frac{\phi(a_{i-1}) - \phi(a_i)}{q_t^{k,i}\sqrt{dt}} \quad \text{for } i = 0, \dots, N+1. \tag{8}$$

while $E_t^{k,i;f} = \eta_t^{k,i;f} = 0$ for $f = 2, \dots, d$. More generally, $\eta_s^{k,i;1} = E_s^{k,i;1}/q_s^{k,i}$ and the other factor loadings remain zero. Consequently,

$$S_s = K^\top q_s^k = \mathbb{E}[S_{T_k} \mid \mathcal{F}_s], \qquad dS_s = K^\top E_s^{k;1}\, dW_s^1,$$

so spot and the front density are driven by the same Brownian factor. At expiry, q_s^k converges to the point mass associated with the sampled strike. The displayed coefficient $E_t^{k,i;1}$ is only the initial value of the exact, state- and time-dependent coefficient $E_s^{k,i;1}$; holding it constant over the whole interval would instead be an Euler approximation. Likewise, sampling S_{T_k} all at once without progressively revealing W^1 produces a direct jump to the terminal point mass rather than a continuous density diffusion.

Case $t \ll T_k$

The same construction gives a natural exact transition over any step which does not cross the front expiry. Let

$$h := T_k - t, \qquad 0 < \delta \le h, \qquad \Delta W_t^1 := W_{t+\delta}^1 - W_t^1.$$

Starting from q_t^k, define Q_i and a_i as above. After drawing $\Delta W_t^1 \sim N(0, \delta)$, set

$$q_{t+\delta}^{k,i} = \Phi\left(\frac{\sqrt{h}\, a_i - \Delta W_t^1}{\sqrt{h - \delta}}\right) - \Phi\left(\frac{\sqrt{h}\, a_{i-1} - \Delta W_t^1}{\sqrt{h - \delta}}\right), \qquad \delta < h. \tag{9}$$

The corresponding spot is not sampled independently but is defined by

$$S_{t+\delta} := K^\top q_{t+\delta}^k = \mathbb{E}[S_{T_k} \mid \mathcal{F}_{t+\delta}]. \tag{10}$$

Consequently

$$\mathbb{E}_t[q_{t+\delta}^k] = q_t^k, \qquad \mathbb{E}_t[S_{t+\delta}] = S_t$$

exactly. In general $S_{t+\delta}$ is not a grid strike; only at T_k does the density collapse to a unit vector and spot become the corresponding strike.

This update is recursively consistent. Indeed, the Gaussian boundaries calculated from the cumulative probabilities of $q_{t+\delta}^k$ are precisely the residual boundaries appearing in (9). The construction may therefore be restarted from $q_{t+\delta}^k$ without changing the joint law. It is a simplex-valued information bridge: conditional on the exact terminal Gaussian variable, W^1 is a Brownian bridge; conditional only on the terminal strike, it is Brownian motion conditioned to terminate in an interval, or equivalently a mixture of Brownian bridges.

Sampling

A stable simulation should not advance spot, the front density and the future densities as separate Euler equations. On a time grid $t_n < t_{n+1}$ with $\delta_n = t_{n+1} - t_n$, one may proceed as follows.

First draw one Brownian vector $\Delta W_n \in \mathbb{R}^d$, using its first coordinate both in (9) and in every DLV coordinate with a non-zero loading $\xi^{j,i;1}$. This preserves the specified leverage. Update the front density by (9), and calculate spot from (10). This part of the update has no time-step bias, preserves positivity and unit mass, and is exactly martingale preserving.

For the DLV squared it is useful to work with $X_t^{j,i} := \log \Sigma_t^{j,i}$. Its Itô dynamics are

$$dX_t^{j,i} = \left(\mu_t^{j,i} - \frac{1}{2} \sum_{f=1}^{d} (\xi_t^{j,i;f})^2 \right) dt + \sum_{f=1}^{d} \xi_t^{j,i;f} \, dW_t^f .$$

Even the frozen-coefficient exponential update

$$\Sigma_{n+1}^{j,i} = \Sigma_n^{j,i} \exp\left\{ \left(\mu_n^{j,i} - \frac{1}{2} \sum_{f} (\xi_n^{j,i;f})^2 \right) \delta_n + \sum_{f} \xi_n^{j,i;f} \Delta W_n^f \right\} \tag{11}$$

is preferable to Euler integration of Σ: it cannot create a negative local variance. For state-dependent coefficients, a genuine Itô weak-order-two stochastic Runge–Kutta method in the log variables reduces the remaining time-step bias.

After updating Σ^j reconstruct the densities expiry by expiry by solving the tridiagonal systems

$$q_{n+1}^j = \left(I - \Omega_j S_{n+1}^j \right)^{-1} q_{n+1}^{j-1}, \qquad j = k+1, \ldots, M. \tag{12}$$

This enforces the DLV relation at every simulated state. Because the matrix is an M-matrix for non-negative Σ, the solve preserves positivity.

The drift condition (5) removes the infinitesimal drift of every density coordinate. It does not, by itself, make a generic finite-step discretisation exactly unbiased, since the map $\Sigma^j \mapsto (M^j)^{-1} q^{j-1}$ is nonlinear. Thus (11), or any ordinary higher-order SDE scheme, retains a time-discretisation bias which should vanish under step refinement.

Discrete Sampling

If exact martingality under a chosen finite-step simulation law is required, it can instead be imposed directly. Choose symmetric discrete shock branches $\Delta W_{n,b}$ with probabilities w_b and construct the front-density branches so that

$$\sum_b w_b q_{n+1,b}^k = q_n^k \, .$$

Parameterise each positive DLV update with an unknown drift correction $c_n^j \in \mathbb{R}^N$. For example, replace the log drift in (11) by $c_n^{j,i}$. For each expiry in turn, solve the nonlinear system

$$\sum_b w_b q_{n+1,b}^j(c_n^j) = q_n^j, \qquad q_{n+1,b}^j(c_n^j) = \left(I - \Omega_j S_{n+1,b}^j(c_n^j) \right)^{-1} q_{n+1,b}^{j-1}. \tag{13}$$

Only N coordinates are independent because mass and first moment are already preserved, matching the N drift corrections. Initialising the nonlinear solver with the continuous-time drift (5) gives a stable finite-step correction. Equation (13) makes the densities exactly martingales under the discrete branch law; symmetric or antithetic branches reduce variance but do not replace this condition.

An option market model reverses the usual direction of construction: instead of specifying a process for the underlying and deriving option prices from it, one models the code-book of prices directly, so that a calibration is a change of coordinates rather than an inverse problem. Two conditions must then be maintained at once. The state must remain free of *static* arbitrage, and its dynamics must admit one equivalent local martingale measure common to the underlying and to every traded option. The second is the option analogue of an HJM drift restriction: the volatility structure may be specified freely, and the drift is then determined. The lattice here is *finite*, finitely many strikes and finitely many expiry dates, which is what is traded.

What is already known. Schweizer and Wissel treat the problem in the continuum in two companion papers, one in each direction of the surface: [18] varies maturity at a single strike, and [17] varies strike at a single maturity. In each the code-book is chosen so that its positivity encodes absence of arbitrage, the drift restriction is derived, and existence is proved. The multi-strike multi-maturity case, which is ours, is stated as open in [17], with the expectation that a new parametrisation will be needed. Carmona and Nadtochiy [10] give a continuum local-volatility code-book with a drift restriction, but no existence theorem. On a finite lattice, and answering the multi-maturity question, stands Wissel [21]; his construction is the one this paper is closest to, and it is described separately below.

Modern approaches. The question was reopened from the data side, by the scarcity of option-market histories for training and evaluating hedging strategies [5]. Cohen, Reisinger and Wang [12] build neural stochastic differential equations on a floating lattice and penalise a least-squares residual in place of the drift restriction. [19] simulate an equity option book with a generative adversarial network whose state is the discrete local variance grid compressed by principal components, so that static arbitrage is absent by positivity of that grid; that is the code-book choice this paper inherits, and [20] extend it to a multi-asset universe. [7] decode statically arbitrage-free surfaces from a low-dimensional code, fit a linear state process to it, and test for dynamic arbitrage after the fact, reporting it in 12.5% of states at zero transaction cost.

None of the four enforces the drift restriction exactly, and that is where this paper starts: the cross-sectional constraints govern each date's surface and its terminal payoff cone, and leave the transition of a fixed contract between two dates unconstrained. The choice of grid is part of the same point. All four index their nodes relative to the spot rather than in fixed cash terms, and on such a grid a node is not one contract through time, so there is no single instrument whose price is required to be a martingale and nothing for an HJM-type restriction to constrain. Fixing the strikes in cash is what buys the restriction here.

The closest prior work. Wissel [21] works on a finite lattice of fixed cash strikes and fixed expiry dates, parametrises it by a price level together with positive local implied volatilities, derives the no-arbitrage drift restrictions, and proves constructive strong existence and uniqueness for a class of coefficients. Behind the front expiry his local implied variance is *exactly* the backward discrete local variance used here, an algebraic identity on the grid, and his drift restriction is, after a change of measure, exactly the one we state in Theorem 3.2. We claim no priority for it. That working paper appears never to have been published in archival form.

What this paper does

We keep the same lattice and change the chart: the state is the vector of gridded *densities* together with the discrete local variances, rather than prices together with quantiles of digitals. Three consequences follow, and they are the contributions.

1. **The spot is a read-off, and it settles exactly.** The spot is $S_t = K^\top q_t^j$ at every expiry j (Proposition 3.3), an identity of the chart holding at all t. At the front expiry Proposition 4.2 exhibits a family of front volatilities under which q^1 collapses onto a single node, so the spot lands on a strike and every gridded contract settles at its exact payoff (Section 4.2). That pinning is a property of the volatility chosen and not of the chart: by Remark 4.3 the terminal states the chart admits carry every spot in the grid range, whereas in [21] the pinning follows from the chart whatever the coefficients. We do not construct dynamics reaching an off-lattice terminal spot; that is left as future work.

2. **Existence, with no condition on the wings.** For every bounded and locally Lipschitz volatility structure b and any number of driving factors, the model has a unique strong solution on $[0, T]$ for every $T < T_1$, positive by construction; it extends to T_1 (Theorem 5.6(ii)) and, restarting at each expiry with the settling front, to the whole ladder $[0, T_M]$, unique given that front (Corollary 5.8). The quantity that must be controlled is $\eta = \psi/q$, the density volatility relative to the density, and its growth across expiries is bounded by the grid alone: Theorem 5.4 gives $\Theta_j \le N + 1$ uniformly in the state and in the volatility structure. *Wissel [21] must assume a bound of this kind; here it is a theorem.* The front expiry costs one hypothesis, and it is a mild one: η^1 must be integrable in t, not bounded, because η^j is affine in η^1 and an Itô equation needs an integrable drift rather than a bounded one. The front volatility used here diverges at rate $(T_1 - t)^{-p}$, and the two demands made of p leave a window: $p < 1$ is what Assumption 5.5 asks, while $p \ge \frac{1}{2}$ is what makes the front density collapse, so that every gridded contract settles at its exact payoff (Proposition 4.2).

3. **The drift is pointwise, and the count works out.** In density coordinates the drift equation is $N + 2$ conditions on N unknowns, and is consistent because the left null space of $\widehat{G}_j$ is exactly $\operatorname{span}\{\mathbf{1}, K\}$, conservation of mass and of the spot (Lemma 2.4). The state-independent factor then cancels from both sides and the drift is available nodewise, $\alpha^{j,i} = -\langle \eta^{j,i}, \beta^{j,i} \rangle$, with no matrix inversion and no rank condition.

Scope, and what is not claimed

The over-determination of the third contribution is not intrinsic: it is created by the choice of density coordinates and removed by the same choice, and in price coordinates the system is square and the question does not arise.

The existence statement is for a fixed finite grid. The constant is $(M-1)(N+1)$ and depends on the grid. The one hypothesis, Assumption 5.5, is a condition on the front volatility and not on the state, and it is sufficient rather than necessary.

2 The code-book

We assume deterministic forwards and discount factors, which reduces to a driftless spot with unit discount factors, cf. [4]. The spot at $t = 0$ is S_0.

The approach we are presenting relies on a combination of a time-and-state backbone process X and an additional smoothing process Y, following the [6] approach to smoothing. However, contrary to that model here we present a full stochastic local volatility model. The weight of the model is carried by the former and echos ideas of [21].

We will define the backbone process on a grid of fixed expiries $0 = T_0 < T_1 < \cdots < T_M$ and strikes $0 < K^0 < K^1 < \cdots < S_0 < \cdots < K^N < K^{N+1}$. We will use two time indices: t will denote model time and j will index fixed expires. We will use i to index strikes Contrary to [6] and similar to [21] expiries and strikes are not floating; they are fixed. The model will evolve along the grid. Specifically, at time t the future expiries in the model are those T_j such that $T_j > t$. We assume that the largest strike spacing is $h := \max_i(K^{i+1} - K^i)$.

2.1 Discrete Local Volatility Backbone

At time t we denote by $\Sigma_t^j = (\Sigma_t^{j,i})_{i=1,\dots,N}$ the squared discrete local volatility at the interior nodes $i = 1, \dots, N$ for expiry $T_j > t$. We let $\lambda = \log \Sigma$ throughout. Define

$$\omega^k := \frac{(K^k)^2}{K^{k+1} - K^{k-1}}, \qquad \iota^{k,-} := \frac{\omega^k}{K^k - K^{k-1}}, \qquad \iota^{k,+} := \frac{\omega^k}{K^{k+1} - K^k}, \qquad k = 1, \dots, N, \tag{14}$$

and then the tridiagonal matrix $A_j(\Sigma)$ with rows

$$\big(A_j(\Sigma)q\big)_i = \begin{cases} q^0 - \Delta_j \Sigma^1 \iota^{1,-} q^1, & i = 0, \\ -\Delta_j \Sigma^{i-1} \iota^{i-1,+} q^{i-1} + \big(1 + \Delta_j \Sigma^i (\iota^{i,-} + \iota^{i,+})\big) q^i - \Delta_j \Sigma^{i+1} \iota^{i+1,-} q^{i+1}, & 1 \le i \le N, \\ -\Delta_j \Sigma^N \iota^{N,+} q^N + q^{N+1}, & i = N+1. \end{cases} \tag{15}$$

(here q^i denotes the i-th component of a generic $q \in \mathbb{R}^{N+2}$). Up to notation this is the implicit forward operator of [9].

Let now X_t be the current spot price at time t for the backbone process X and let $q_t^0 \in \mathbb{R}^{N+2}$ the density

$$q_t^{0,i} = \frac{(X_t - K^{i-1})^+ - (X_t - K^i)^+}{K^i - K^{i-1}} - \frac{(X_t - K^i)^+ - (X_t - K^{i+1})^+}{K^{i+1} - K^i}$$

Then, the family $q_t = (q_t^j)_{j=0,\dots,M}$ starting in q_t^0 is then given iteratively as

$$A_j(\Sigma_t^j)\, q_t^j = q_t^{j-1} \tag{16}$$

Note that in (15) the entry multiplying $q_t^{i\pm1}$ carries $\Sigma_t^{i\pm1}$ and not Σ_t^i. The following was shown in [9]:

Theorem 2.1. *The family is then $q = (q_t^{j,i})$ with $q_t^j \in \mathbb{R}^{N+2}$ us a discrete risk-neutral density on the grid (T, K) in the sense that*

1. $q_t^{j,i} \ge 0$
2. $\sum_{i=0}^{N+1} q_t^{j,i} = 1$
3. $\sum_{i=0}^{N+1} K^i q_t^{j,i} = X_0$.

This means there is a discrete time-and-space process $X_t = (X_t^j)_{j=0,\dots,M}$ which takes values only in (T, K). Note that the model-time index here is j. Call prices for X_t for arbitrary $(\tau, k) \in \mathbb{R}^2_{\geq 0}$ are given by the linear functional

$$C_t(k,\tau) = \mathbb{E}\left[\left(X_t^j - k\right)^+\right] = \sum_{i=0}^{N+1} (K^i - k)^+ q_t^{j,i} \ . \tag{17}$$

with $j = \max\{j : T_j \leq \tau\}$.

Because (15) is affine in Σ we may write

$$A_j(\Sigma)\, q \ = \ q + G_j(q)\, \Sigma, \qquad G_j(q) \in \mathbb{R}^{(N+2)\times N}, \tag{18}$$

so that $G_j(q)$ collects the terms carrying Σ. The map G_j is linear in q and $G_j(q)\Sigma$ is linear in each argument separately; we use both readings. The following elementary observation is used constantly.

Lemma 2.2 (Factorisation)**.** *Column k of $G_j(q)$ involves q only through q^k. Consequently*

$$G_j(q) \ = \ \widehat{G}_j \,\mathrm{diag}(q^1, \dots, q^N), \tag{19}$$

where $\widehat{G}_j \in \mathbb{R}^{(N+2)\times N}$ has k-th column $\Delta_j\big(-\iota^{k,-}e_{k-1} + (\iota^{k,-} + \iota^{k,+})e_k - \iota^{k,+}e_{k+1}\big)$. Moreover $\widehat{G}_j$ has rank N, hence is injective.

Proof. Immediate from (15), whose terms carrying Σ^k are all proportional to q^k. For injectivity, suppose $\widehat{G}_j c = 0$ and let k be the smallest index with $c_k \neq 0$. Row $k-1$ receives a contribution from column k' only for $k' \in \{k-2, k-1, k\}$, and the first two carry $c_{k'} = 0$. So row $k-1$ reads $-\Delta_j \iota^{k,-} c_k = 0$, forcing $c_k = 0$, a contradiction. □

Note that $\widehat{G}_j$ depends only on the grid and on Δ_j, not on the state; the state enters (19) only through the diagonal factor.

Proposition 2.3 (The code-book)**.** *Fix the front density q^1, a probability vector on the grid with mean S_0, and let $j = 2, \dots, M$.*

(i) Given densities q^{j-1} and q^j with $q^j > 0$, the discrete local variance is uniquely determined by the linear system

$$G_j(q^j)\, \Sigma^j \ = \ q^{j-1} - q^j. \tag{20}$$

(ii) Conversely, given $\Sigma^j > 0$, the vector $q^j := A_j(\Sigma^j)^{-1} q^{j-1}$ is again a probability vector with mean S_0, and the gridded prices $(\pi^i)^\top q^j$ admit no static arbitrage.

(iii) A gridded surface is free of static arbitrage, butterflies and calendar spreads together, if and only if it arises as in (ii) *from some $\Sigma \geq 0$, and it lies in the* interior *of the arbitrage-free set if and only if $\Sigma > 0$.*

Proof. (i) is (18), with uniqueness from Lemma 2.2. For (ii), $A_j(\Sigma^j)^{-1}$ is non-negative with unit column sums and inherits the martingale property [9], attributed there to Andreasen and Huge [1]; non-negativity excludes butterfly arbitrage, and q^j is then a mean-preserving spread of q^{j-1}, so Jensen's inequality excludes calendar arbitrage. (iii) is [9], after [3]: absence of arbitrage is equivalent to non-negative Gamma and Backward-Theta at every node, and Σ is their ratio. Both conditions are nodewise, and $\Sigma^{j,i} = 0$ exactly when Backward-Theta vanishes. □

What everything else rests on is that the admissible set is a product of half-lines, one per node, so it has no faces in the sense that a polytope does, and $\lambda = \log \Sigma$ maps its interior diffeomorphically onto $\mathbb{R}^{N(M-1)}$.

2.2 Two conservation laws

The whole structure of the drift restriction follows from one observation about the left null space of G_j. Both identities below are used by [9], where they hold by construction for the backward operator. What is not there, because nothing in that paper needs it, is that $\operatorname{span}\{\mathbf{1}, K\}$ is the *entire* left null space. That is a rank statement, and it is what makes the drift system of Section 3 square.

Lemma 2.4 (Left null space)**.** *For every q and every j,*

$$\mathbf{1}^\top G_j(q) = 0 \qquad \text{and} \qquad K^\top G_j(q) = 0,$$

where $\mathbf{1} = (1, \dots, 1)$ and $K = (K^0, \dots, K^{N+1})$. Consequently $A_j(\Sigma)$ conserves both total mass and the spot, for any Σ. Combined with Lemma 2.2, the left null space of $\widehat{G}_j$ is exactly $\operatorname{span}\{\mathbf{1}, K\}$.

Proof. Fix an interior index k and collect the three entries of $G_j(q)$ in column k: from (15) they are $\Delta_j(\iota^{k,-} + \iota^{k,+})q^k$ in row k, $-\Delta_j \iota^{k,+} q^k$ in row $k+1$, and $-\Delta_j \iota^{k,-} q^k$ in row $k-1$. Their sum is 0, giving the first identity. Weighting by the node values,

$$\begin{aligned}\Delta_j q^k \Big[K^k\big(\iota^{k,-} + \iota^{k,+}\big) - K^{k+1}\iota^{k,+} - K^{k-1}\iota^{k,-}\Big] &= \Delta_j q^k \Big[\iota^{k,-}(K^k - K^{k-1}) - \iota^{k,+}(K^{k+1} - K^k)\Big] \\ &= \Delta_j q^k \big[\omega^k - \omega^k\big] = 0\end{aligned}$$

by (14), giving the second. The conservation statements follow from (18): $\mathbf{1}^\top A_j(\Sigma) q = \mathbf{1}^\top q$ and $K^\top A_j(\Sigma) q = K^\top q$. Since $\widehat{G}_j$ has rank N and lives in $\mathbb{R}^{(N+2)\times N}$, its left null space is two-dimensional, hence exactly $\operatorname{span}\{\mathbf{1}, K\}$. □

Remark 2.5 (Counting). For $q > 0$, $G_j(q)$ has $N+2$ rows and N columns and rank N. Hence the $N+2$ martingale conditions per expiry contain exactly two redundancies, conservation of mass and of the spot, and the remaining N conditions match the N unknowns of Σ^j. The system is square on the quotient.

3 The drift restriction

Theorem 3.1 (Martingale property)**.** *Assume that there is a measure $\mathbb{Q}$ such that each $q_t^{j,i}$ is a martingale. Then X_t and all options written on it are martingales.*

Under $\mathbb{Q}$ every gridded instrument price is a martingale in t. Since the gridded payoffs π^i together with the unit payoff span $\mathbb{R}^{N+2}$, and $\mathbf{1}^\top q^j = 1$, this is equivalent to each density vector being a martingale,

$$\mathrm{d}q_t^j = \psi_t^j \,\mathrm{d}W_t, \qquad \psi_t^j \in \mathbb{R}^{(N+2)\times d}. \tag{21}$$

This is the only assumption behind the drift restriction, and it carries no regularity. Each q^j is a fixed linear combination of traded butterflies, so (21) is the martingale representation theorem applied to traded prices, and nothing has to commute with a stochastic integral. This is the substantive advantage of a discrete code-book over its continuum analogue, where the corresponding step requires differentiating a stochastic field in strike and maturity and must be licensed by standing regularity conditions. Theorem 3.2 below needs only $q^j > 0$.

Theorem 3.2 (Drift restriction)**.** *Let* $\mathrm{d}\Sigma_t^j = \alpha_t^j \mathrm{d}t + \beta_t^j \mathrm{d}W_t$ *and* $q^j > 0$*. Under Assumption 3.1, for each* $j = 2, \dots, M$*,*

$$A_j(\Sigma^j)\,\psi^j \;=\; \psi^{j-1} - G_j(q^j)\,\beta^j, \tag{22}$$

marched forward in j *from* ψ^1*, the volatility of the front density, which is given by its chart (Section 4). With the relative density volatility*

$$\eta^{j,i} \;:=\; \frac{\psi^{j,i}}{q^{j,i}}, \qquad i = 0, \dots, N+1, \tag{23}$$

the drift is determined node by node,

$$\alpha^{j,i} \;=\; -\langle \eta^{j,i}, \beta^{j,i} \rangle \qquad \Longleftrightarrow \qquad \mathrm{d}\lambda^{j,i} = \Big[-\langle \eta^{j,i}, b^{j,i} \rangle - \tfrac{1}{2} |b^{j,i}|^2 \Big] \mathrm{d}t + b^{j,i}\,\mathrm{d}W_t, \tag{24}$$

where $\lambda = \log \Sigma$ *and* $b = \beta / \Sigma$*.*

Proof. By Lemma 2.2, $G_j(q)\Sigma = \widehat{G}_j(\Sigma \circ q)$, where $\Sigma \circ q := (\Sigma^i q^i)_{i=1}^N$ is the entrywise product on the interior nodes. The code-book therefore reads

$$q_t^j + \widehat{G}_j\, p_t^j = q_t^{j-1}, \qquad p^{j,i} := \Sigma^{j,i} q^{j,i}, \quad i = 1, \dots, N,$$

with $\widehat{G}_j$ constant. Itô's product rule for the scalar processes $\Sigma^{j,i}$ and $q^{j,i}$ gives

$$\mathrm{d}p^{j,i} = \Big[q^{j,i} \alpha^{j,i} + \langle \psi^{j,i}, \beta^{j,i} \rangle \Big] \mathrm{d}t + \big(\Sigma^{j,i} \psi^{j,i} + q^{j,i} \beta^{j,i} \big) \mathrm{d}W_t, \tag{25}$$

with $\psi^{j,i}, \beta^{j,i} \in \mathbb{R}^{1 \times d}$ the rows at node i. Hence $\mathrm{d}q^j + \widehat{G}_j\,\mathrm{d}p^j = \mathrm{d}q^{j-1}$, and by uniqueness of the decomposition of a continuous semimartingale the $\mathrm{d}W$ and $\mathrm{d}t$ terms match separately. The $\mathrm{d}W$ terms give $\psi^j + \widehat{G}_j \big(\Sigma^j \circ \psi^j + q^j \circ \beta^j \big) = \psi^{j-1}$, factor by factor, which is (22) by (18). Since q^j and q^{j-1} are driftless by Assumption 3.1, the $\mathrm{d}t$ terms give

$$\widehat{G}_j \Big(q^j \circ \alpha^j + \textstyle\sum_f \psi^{j,\cdot,f} \circ \beta^{j,\cdot,f} \Big) = 0.$$

Since $\widehat{G}_j$ is injective (Lemma 2.2) the bracket vanishes, $q^{j,i}\alpha^{j,i} + \langle \psi^{j,i}, \beta^{j,i} \rangle = 0$ node by node, which is (24) after dividing by $q^{j,i} > 0$. The logarithmic form is Itô applied to $\lambda = \log \Sigma$, using $\alpha / \Sigma = -\langle \eta, b \rangle$. □

Only the interior components of η enter (24), since Σ lives on the interior nodes; the boundary components are needed in Section 5.

Equation (22) is not ours. Compared with the pricing recursion $A_j(\Sigma^j) q^j = q^{j-1}$ it is *the same operator with one added source*, the structure of Carmona and Nadtochiy [10]. The drift restriction above is theirs in the continuum, and Wissel's [21] on this lattice; we claim no priority for either.

What the fixed lattice changes is threefold. Their restriction carries a term in the derivative of the code-book along time to maturity; a lattice of fixed expiry dates has no such derivative, and here that term is absent. The relation is algebraic rather than differential, so neither the high-order regularity their statement requires nor the singular behaviour as time to maturity vanishes that it is built to accommodate can arise here, the front slab having positive width. And the discrete system is over-determined, $N + 2$ conditions on N unknowns, which is what the counting of Remark 2.5 squares.

Two features of (24) are worth drawing out. The first is its shape. It is the exact analogue of the HJM restriction $\alpha = \sigma \int \sigma$: the drift of the log discrete local variance is minus the covariance of its own volatility with the volatility of the log density, plus an Itô convexity term. Where HJM's derived object is an integral of the volatility across maturities, ours is η, obtained from b by the single forward recursion (22). The second is the cost. Once η is known no matrix is inverted: no Jacobian of the pricing map is ever formed and no rank condition on it appears. And since ψ is affine in β and α is bilinear in (β, ψ), the drift is a quadratic polynomial in β, as in HJM; Theorem 5.4 is what rules out the corresponding finite-time explosion.

The driftless point is the classical model: if $\beta \equiv 0$ and $\psi^1 = 0$ then $\psi \equiv 0$ and $\alpha \equiv 0$, so the discrete local variance surface is frozen, and with $\psi^1 \neq 0$ the densities still move, driven by the front alone, with $\alpha \equiv 0$ persisting.

Proposition 3.3 (Spot consistency)**.** *The spot is $S_t = K^\top q_t^j$, the same for every j by Lemma 2.4. Since $\mathbf{1}^\top A_j = \mathbf{1}^\top$, $K^\top A_j = K^\top$ and $\mathbf{1}^\top \widehat{G}_j = K^\top \widehat{G}_j = 0$, equation (22) gives*

$$\mathbf{1}^\top \psi^j = \mathbf{1}^\top \psi^1 = 0 \qquad \text{and} \qquad K^\top \psi^j = K^\top \psi^1 = \sigma_t^S S_t \quad \text{for every } j.$$

So the spot's volatility is the single row $K^\top \psi^1 \in \mathbb{R}^{1\times d}$, it is automatically the same at every expiry, and no further condition links the spot to the surface. Taking $\psi^1 = 0$ freezes the spot and recovers $K^\top \psi^j = 0$.

Proposition 3.3 is the discrete form of Carmona and Nadtochiy's spot consistency condition: the spot and its volatility are both *read off* the surface rather than modelled, $S = K^\top q^j$ and $\sigma^S S = K^\top \psi^j$ for any j. If $K^0 = 0$, the read-off is the zero-strike call, $S_t = C_t(0, T_j)$ for every j, as in [21] and in the multi-strike model of [17].

The drift restriction is used below through the following chain, which is what a model actually evaluates:

$$(\psi^1, b) \xrightarrow{(22)} \psi^2, \ldots, \psi^M \xrightarrow{\eta = \psi/q} \alpha, \tag{26}$$

a forward march of d tridiagonal solves per expiry, $O(NMd)$ in total. The volatility structure is the only input; the drift is computed, never fitted.

4 The spot and the front expiry

The spot is not a state variable. By Proposition 3.3 it is the read-off $S_t = K^\top q_t^j$, the same scalar at every expiry. What has to be specified is the front density q^1, the initial condition of the recursion of Proposition 2.3; everything at $j \geq 2$ follows from it and from the volatility structure.

4.1 The front chart

The front density must be strictly positive and sum to one. Both are automatic in the chart

$$q^{1,i} = \frac{e^{z^i}}{\sum_k e^{z^k}}, \qquad \mathrm{d}z = m\,\mathrm{d}t + \sum_{f=1}^{d} \nu^f\,\mathrm{d}W^f, \tag{27}$$

with $z \in \mathbb{R}^{N+2}$ unconstrained. The chart is invariant under $z \mapsto z + c\mathbf{1}$, so a shift of ν along $\mathbf{1}$ leaves q^1 unchanged and only the q^1-centred part of ν is identified; write $\mathrm{dev}^f := \nu^f - (q^1)^\top \nu^f\,\mathbf{1}$

for it. The input ν is free, and plays the role b plays at $j \geq 2$. Itô's formula for a function of a vector process shows that each $q^{1,i}$ is a martingale exactly when

$$m^i = -\tfrac{1}{2} \sum_f \big(\mathrm{dev}^{i,f}\big)^2 \tag{28}$$

up to a multiple of $\mathbf{1}$. Furthermore we obtain

$$\psi^{1,\cdot,f} = q^1 \circ \mathrm{dev}^f, \qquad \eta^{1,\cdot,f} = \mathrm{dev}^f, \qquad \mathbf{1}^\top \psi^{1,\cdot,f} = 0, \tag{29}$$

with $\circ$ the entrywise product. The same computation appears in [13] and as the stochastic replicator equation [15].

Remark 4.1 (Resolution at the front expiry). Among grid measures with mean S the least dispersed is the pair of nodes straddling it, so

$$\mathrm{Var}_{q^1}(K) \;\geq\; (S - K^i)(K^{i+1} - S), \qquad K^i \leq S < K^{i+1}, \tag{30}$$

which is at most $\frac{1}{4}h^2$. The front implied volatility therefore has the floor

$$\frac{\sqrt{(S - K^i)(K^{i+1} - S)}}{S\sqrt{T_1 - t}} \;\leq\; \frac{h}{2S\sqrt{T_1 - t}},$$

largest at the midpoint of a cell and zero when the spot sits on a node, which is exactly what Proposition 4.2 forces at T_1.

4.2 Terminal settlement: pinning the spot to a strike

The chart constrains q^1 at no time $t < T_1$ beyond positivity and mass, so it imposes no terminal condition. Wissel's front parametrisation fixes the terminal state whatever the coefficients [21]; here the terminal state is a modelling choice. We take the front volatility proportional to the distance from the spot, in one factor,

$$\eta_t^{1,i} = \frac{c_t\,(K^i - S_t)}{h}, \qquad c_t = a_1 \tau^{-p} \text{ near } T_1, \quad \tau = T_1 - t,\ a_1 > 0. \tag{31}$$

Only the behaviour near T_1 matters: on $[0, T_1 - \varepsilon]$ the front volatility may be any bounded, locally Lipschitz choice, and (31) need only hold on $[T_1 - \varepsilon, T_1)$. It is already centred, since $\sum_i q^{1,i}(K^i - S_t) = 0$. Its coefficients are polynomial in q^1 and bounded on $[0, T]$ for every $T < T_1$, so there is a unique strong solution on $[0, T_1)$,

$$q_t^{1,i} = q_0^{1,i} \exp\Big(\int_0^t \eta^{1,i}\,\mathrm{d}W - \tfrac{1}{2}\int_0^t (\eta^{1,i})^2 \mathrm{d}s \Big), \qquad \|\eta_t^1\|_\infty \leq c_t\, w/h, \quad w := K^{N+1} - K^0. \tag{32}$$

Proposition 4.2 (The front collapses onto a node, and settles exactly). *If $p \geq \frac{1}{2}$ then $q_{T_1}^1 := \lim_{t \uparrow T_1} q_t^1 = e_k$ for a random index k, almost surely, so $S_{T_1} = K^k$ and*

$$\sum_i q_{T_1}^{1,i} (K^i - K^n)^+ = (S_{T_1} - K^n)^+ \qquad \textit{for every } n. \tag{33}$$

If $p < 1$ then Assumption 5.5 holds on $[0, T_1]$. The window $p \in [\frac{1}{2}, 1)$ gives both.

Proof. Each $q^{1,i}$ is a bounded martingale, so $q^1_t \to q^1_{T_1}$ almost surely, and the variance $V_t := \sum_i q^{1,i}_t (K^i - S_t)^2$ converges to that of $q^1_{T_1}$. By (29) and (31) the volatility of $S = K^\top q^1$ is $\sum_i K^i q^{1,i} \eta^{1,i} = c_t V_t / h$ on $[T_1 - \varepsilon, T_1)$. Since S is a martingale with values in $[K^0, K^{N+1}]$,

$$\mathbb{E} \int_{T_1-\varepsilon}^{T_1} \frac{c_t^2 V_t^2}{h^2} \, \mathrm{d}t = \mathbb{E}\big[(S_{T_1} - S_{T_1-\varepsilon})^2\big] \le w^2,$$

so the integral is finite almost surely. On the event $\{V_{T_1} > 0\}$ we would have $V_t \ge V_{T_1}/2$ near T_1, and the integral would diverge, because $\int^{T_1} c_t^2 \, \mathrm{d}t = a_1^2 \int^{T_1} \tau^{-2p} \, \mathrm{d}t = \infty$ for $2p \ge 1$. Hence $V_{T_1} = 0$ almost surely: $q^1_{T_1}$ is a probability vector on distinct points with zero variance, that is e_k with $K^k = S_{T_1}$, and (33) reads $(K^k - K^n)^+ = (S_{T_1} - K^n)^+$. Finally η^1 is bounded on $[0, T_1 - \varepsilon]$ and $\int_{T_1-\varepsilon}^{T_1} \|\eta^1_t\|_\infty \mathrm{d}t \le (w/h) \int_{T_1-\varepsilon}^{T_1} c_t \, \mathrm{d}t < \infty$ when $p < 1$, by (32). □

The terminal spot is therefore a grid point, as in [21]; the difference is that here it is chosen, by (31), and there it is forced by the parametrisation.

Remark 4.3 (What the lattice could carry, and does not). At expiry each butterfly price $q^{j,i}$ is worth its payoff: for $S_{T_j} \in [K^k, K^{k+1}]$ the second difference defining $q^{j,i}$ telescopes to $\varpi(S_{T_j})$, the barycentric weights of the spot on the two nodes straddling it. Since $u \mapsto (u - K^n)^+$ is affine between nodes,

$$\sum_i \varpi^i(s) \, (K^i - K^n)^+ \;=\; (s - K^n)^+ \qquad \text{for every } s \in [K^0, K^{N+1}] \text{ and every model strike } K^n, \tag{34}$$

so $\varpi(s)$ is an admissible terminal state that settles every gridded contract exactly, also for s strictly inside a cell. The chart does not force the pinning; the choice (31) does, because its terminal variance must vanish. A front volatility that drives q^1 to $\varpi(s)$ with s inside a cell is left as future work (Section 7).

5 The wing constant, and existence

The log discrete local variance $\lambda = (\lambda^2, \dots, \lambda^M)$, $\lambda^j = \log \Sigma^j$, ranges over all of $\mathbb{R}^{N(M-1)}$, so positivity is automatic; the front density has its own chart (27). By Theorem 3.2 the model is the stochastic differential equation

$$\mathrm{d}\lambda_t = \Big[-\langle \eta_t, b_t \rangle - \tfrac{1}{2} |b_t|^2 \Big] \mathrm{d}t + b_t \, \mathrm{d}W_t, \tag{35}$$

whose only non-elementary coefficient is $\eta = \psi / q$ from (22) and (23). Its denominator is small in the wings, and the question is whether η can blow up there.

The argument has the structure of Wissel's proof of his Theorem 3.5 [21]: log coordinates, induction over expiries, and local existence followed by a bound that rules out explosion. Its one non-standard ingredient is an a priori bound, independent of the state, on the relative volatility of the calendar spread at the second expiry. Wissel could not obtain this bound in general and imposes his condition (3.24) instead (his Remark 1 following the proof, p. 18); in our coordinates the quantity is $\eta + b$. We obtain it from an exact identity for the volatility recursion and a known inequality for inverses of M-matrices.

5.1 The volatility recursion

Write $P_j := A_j(\Sigma^j)^{-1}$, so that $q^j = P_j q^{j-1}$, and P when j is fixed.

Corollary 5.1 (Positivity, unit column sums, absorbing ends)**.** *$P \geq 0$ entrywise, $\mathbf{1}^\top P = \mathbf{1}^\top$, so $P_{im} \in [0,1]$, and the outer columns of P are e_0 and e_{N+1}.*

Proof. By (15), A_j has non-positive off-diagonal entries, and in each column the diagonal entry exceeds the sum of the moduli of the off-diagonal ones by exactly one; so A_j is a non-singular M-matrix and $P \geq 0$ [2]. The column sums follow from $\mathbf{1}^\top A_j = \mathbf{1}^\top$ (Lemma 2.4), and the outer columns of A_j are e_0 and e_{N+1}. The first two statements are also Theorem 3.3 of [9], attributed there to [1]. □

That is all that is needed about P itself. What we want next is a recursion for η alone: one step per expiry, each adding a controlled amount, so that the bound telescopes along the ladder. Dividing (22) by q^j gives

$$\eta^{j,i} \;=\; \frac{(P\psi^{j-1})_i}{q^{j,i}} \;-\; \frac{\big(P\,G_j(q^j)\beta^j\big)_i}{q^{j,i}}, \tag{36}$$

the first term carrying η^{j-1} and the second carrying b^j.

The first is an average. Writing $\psi^{j-1} = q^{j-1} \circ \eta^{j-1}$, its weights $P_{ik}q^{j-1,k}/q^{j,i}$ are non-negative because $P \geq 0$ and sum to one because $q^j = Pq^{j-1}$, so it contributes at most $\|\eta^{j-1}\|_\infty$; they are the matrix $\mathcal{A}$ of (41).

The source is the difficulty. Since G_j is a second difference, row i carries the neighbouring densities $q^{j,i\pm1}$ as well as $q^{j,i}$, so dividing by $q^{j,i}$ leaves the ratios

$$\frac{q^{j,i-1}}{q^{j,i}} \qquad \text{and} \qquad \frac{q^{j,i+1}}{q^{j,i}}, \tag{37}$$

which are unbounded in the wings, where the density decays fastest; nothing in (22) bounds them.

They are, however, an artefact of writing the row out. Since $\mathbf{1}^\top G_j(q) = 0$ by Lemma 2.4, row i of $G_j(q)\Sigma^j$ is the net outflow of node i, and by (18) that net is

$$G_j(q^j)\,\Sigma^j \;=\; A_j(\Sigma^j)q^j - q^j \;=\; q^{j-1} - q^j, \tag{38}$$

which carries no ratios at all: the individual fluxes are wing quantities, their net is not. For $b^j \equiv c$ this settles the matter, since $\beta^j = c\,\Sigma^j$ and $Pq^{j-1} = q^j$ give

$$\frac{\big(P\,G_j(q^j)\beta^j\big)_i}{q^{j,i}} = c\,\frac{\big(q^j - Pq^j\big)_i}{q^{j,i}} = c\Big(1 - \frac{(Pq^j)_i}{q^{j,i}}\Big),$$

the two ratios of (37) having collapsed into the single propagation ratio $(Pq^j)_i/q^{j,i}$. A general b only modulates the density. Let $E \in \mathbb{R}^{N\times(N+2)}$ restrict to the interior nodes, so $EE^\top = I_N$, and note that $A_j = I + \widehat{G}_j\,\mathrm{diag}(\Sigma^j)E$. Writing $E^\top(q \circ b)$ for the vector equal to $q^i b^i$ at the interior nodes and zero at the outer ones, (19) and the commuting of diagonal matrices give

$$G_j(q)\beta^j = \widehat{G}_j\,\mathrm{diag}(\Sigma^j)E\,E^\top(q \circ b^j) = \big(A_j(\Sigma^j) - I\big)E^\top(q \circ b^j), \tag{39}$$

again the operator's own increment, applied to $E^\top(q \circ b^j)$ in place of q. The proposition is what this amounts to.

Proposition 5.2. *With $b^j = \beta^j/\Sigma^j$ and $\circ$ the entrywise product,*

$$\psi^j \;=\; P\,\psi^{j-1} \;-\; (I-P)\,E^\top\big(q^j \circ b^j\big), \tag{40}$$

equivalently, with the entrywise non-negative

$$\mathcal{A} := \operatorname{diag}(q^j)^{-1} P \operatorname{diag}(q^{j-1}), \qquad \mathcal{B} := \operatorname{diag}(q^j)^{-1} P\,E^\top \operatorname{diag}(Eq^j),$$

$$\eta^j \;=\; \mathcal{A}\,\eta^{j-1} \;+\; \big(\mathcal{B} - E^\top\big) b^j, \tag{41}$$

where $\mathcal{A}\mathbf{1} = \mathbf{1}$, so the transport is an averaging operator, and the i-th row of $\mathcal{B}$ sums to $(Pq^j)_i/q^{j,i} - \mathbf{1}_{\{i\in\{0,N+1\}\}}$.

Proof. By (22), $\psi^j = P\psi^{j-1} - P\,G_j(q^j)\beta^j$. Applying P to (39) at $q = q^j$ and using $PA_j = I$ gives $P\,G_j(q^j)\beta^j = (I-P)E^\top(q^j \circ b^j)$, which is (40). Dividing it by $q^{j,i}$ gives (41): the rows of $\mathcal{A}$ sum to $(Pq^{j-1})_i/q^{j,i} = 1$, and for $\mathcal{B}$ the terms $k \in \{0, N+1\}$ missing from $(Pq^j)_i$ vanish for interior i and equal $q^{j,i}$ at $k = i$ for outer i, by Corollary 5.1. □

So η^j is an average of η^{j-1} plus a source in b^j, and the march across expiries is additive.

5.2 The wing constant

That single surviving ratio, $(Pq^j)_i/q^{j,i}$, is what has to be bounded. Since $q^j = Pq^{j-1}$ it is

$$\frac{(Pq^j)_i}{q^{j,i}} = \frac{(P^2q^{j-1})_i}{(Pq^{j-1})_i}, \tag{42}$$

so what is needed is how much larger than P the square P^2 can be entrywise; and since both sides are linear and non-negative in q^{j-1}, the case $q^{j-1} = e_m$ shows that an entrywise bound is not merely sufficient but necessary. Write

$$R^{j,i} := \mathbf{1}_{\{1\le i\le N\}} + \frac{(Pq^j)_i}{q^{j,i}} - \mathbf{1}_{\{i\in\{0,N+1\}\}}, \qquad \Theta_j := \max_i R^{j,i}, \qquad \Xi_j := \sum_{r=1}^{N} P_{rr} \le N, \tag{43}$$

and $H_j := \max_i|\eta^{j,i}|$. Here $R^{j,i}$ is the row bound of (41) at node i, and Θ_j, the *wing constant*, is its size across the slab. The theorem shows Θ_j is bounded by the grid alone, uniformly in the state; the bound rests on a single inequality for inverses of M-matrices.

Lemma 5.3 (Squared propagator)**.** *For all nodes i, r, m,*

$$P_{ir}\,P_{rm} \;\le\; P_{im}\,P_{rr}, \tag{44}$$

and consequently, for all i, m,

$$\big(P^2\big)_{im} \le \begin{cases} \Xi_j\,P_{im}, & 1 \le i \le N, \\ (1+\Xi_j)\,P_{im}, & i \in \{0, N+1\}. \end{cases} \tag{45}$$

Proof. P is the inverse of an M-matrix by Corollary 5.1, and every inverse M-matrix is a path product matrix, $a_{ij}a_{jk} \le a_{ik}a_{jj}$ for all i, j, k [14, Theorem 4.5]; (44) is the case $(i, j, k) = (i, r, m)$. Summing it over r gives (45): an outer r contributes only if $r = i$, since column r of P is e_r, and then contributes P_{im}. □

Theorem 5.4. *For every $\Sigma \in (0,\infty)^N$,*

$$H_j \;\le\; H_{j-1} + \Theta_j \|b^j\|_\infty, \tag{46}$$

and

$$\Theta_j \le 1 + \Xi_j \le N+1, \qquad \Theta^* := \sup_{\Sigma} \sum_{j=2}^{M} \Theta_j \le (M-1)(N+1). \tag{47}$$

Proof. Since $q^j = Pq^{j-1} \ge 0$, Lemma 5.3 gives $(Pq^j)_i = \sum_m (P^2)_{im} q^{j-1,m} \le \Xi_j\, q^{j,i}$ for interior i and $\le (1+\Xi_j) q^{j,i}$ for outer i, so $R^{j,i} \le 1+\Xi_j$ in both cases. Finally, $\mathcal{A} \ge 0$ with $\mathcal{A}\mathbf{1} = \mathbf{1}$ gives $\|\mathcal{A}\eta^{j-1}\|_\infty \le H_{j-1}$, while row i of $(\mathcal{B} - E^\top) b^j$ has modulus at most $\|b^j\|_\infty R^{j,i}$ by the row sums of Proposition 5.2. □

5.3 Existence

Iterating (46) from the front slice,

$$\sup_{j,i} |\eta_t^{j,i}| \;\le\; \|\eta_t^1\|_\infty + (M-1)(N+1)\, \|b\|_\infty, \tag{48}$$

and iterating (40) shows that η^j is affine in η^1, with coefficients depending on λ only through $P_2, \dots, P_j$ and $q^2, \dots, q^j$. The only t on the right of (48) is inside η_t^1, which enters linearly, so it has to be integrable in time rather than bounded.

Assumption 5.5 (Integrable front volatility)**.** For a horizon $T \le T_1$,

$$\int_0^T \|\eta_t^1\|_\infty \,\mathrm{d}t \;<\; \infty \qquad \text{almost surely.} \tag{49}$$

Theorem 5.6 (Existence)**.** *Let the front volatility $\nu = \nu(t,z)$ in (27) and the volatility $b = b(t,z,\lambda)$ be locally Lipschitz in (t,z,λ) on $[0,T_1)$, with $|b| \le B$, and ν bounded on $[0,T]$ for every $T < T_1$.*

(i) For every $T < T_1$ the front chart (27)–(28) together with (35) has a unique strong solution (z,λ) on $[0,T]$, and

$$0 \;<\; \inf_{t\le T} \min_{j,i} \Sigma_t^{j,i} \;\le\; \sup_{t\le T} \max_{j,i} \Sigma_t^{j,i} \;<\; \infty \qquad \textit{almost surely.} \tag{50}$$

(ii) If moreover Assumption 5.5 holds with $T = T_1$, then λ_t has a finite limit λ_{T_1} as $t \uparrow T_1$, and (50) holds with $T = T_1$.

No condition is imposed on Σ, on the wings, or on the number of factors.

Proof. (i) Fix $T < T_1$ and let B_ν bound $|\nu|$ on $[0,T]$. The coefficients are locally Lipschitz: $q^1 = \mathrm{softmax}(z)$ and $\Sigma = e^\lambda$ are positive, so each A_j is a non-singular M-matrix and everything except ν and b is smooth in (z,λ). Adding time as a state variable to cover the time dependence, [16, Theorem V.38, p. 303] gives a unique solution up to an explosion time ζ, with $\limsup_{t\to\zeta} \|(z_t,\lambda_t)\| = \infty$ on $\{\zeta < \infty\}$. The drift of λ is bounded by (48), since $\|\eta^1\|_\infty = \|\mathrm{dev}\|_\infty \le 2B_\nu$, and the drift m of z is bounded by $2B_\nu^2$; let D bound both. On $[0, \zeta\wedge T)$ the drift part of (z,λ) is therefore at most DT, and the noise part, having integrand bounded by $B_\nu + B$, is a martingale bounded in L^2, hence convergent as $t \to \zeta\wedge T$. So (z,λ) stays bounded

on $[0, \zeta \wedge T)$, which rules out $\zeta \le T$. The solution exists on $[0,T]$, λ is bounded there, and (50) follows from $\Sigma = e^{\lambda}$.

(ii) By (i) the solution exists on $[0,T_1)$. By (48) the drift of λ is bounded by $g(t) := \big(\|\eta_t^1\|_\infty + (M-1)(N+1)B\big)B + \frac{1}{2}B^2$, which is integrable on $[0,T_1)$ by Assumption 5.5, so the drift part of λ converges as $t \uparrow T_1$. The noise part $\int_0^t b\,\mathrm{d}W$ has integrand bounded by B, so it is a martingale bounded in L^2 on $[0,T_1)$ and converges as well. Hence λ_t has a finite limit, λ is bounded on $[0,T_1]$, and (50) holds with $T = T_1$. Nothing is claimed for z, which may run off at T_1 as the front collapses. □

The settling front (31) is bounded on every $[0,T]$ with $T < T_1$ by (32), and satisfies Assumption 5.5 with $T = T_1$ for $p < 1$ (Proposition 4.2). So for $p \in [\frac{1}{2}, 1)$ the code-book reaches its first expiry with Σ bounded away from 0 and ∞ and every gridded contract settling at its payoff.

5.4 Beyond the first expiry

At T_1 the front expires and the next expiry becomes the front. The settling of Proposition 4.2 makes this switch clean: the spot sits on a node, and the new front density is a column of the propagator. Indeed, if $q^l_{T_l} = e_k$ then the code-book $A_{l+1}(\Sigma^{l+1})\,q^{l+1} = q^l$ gives

$$q^{l+1}_{T_l} = P_{l+1}\,e_k, \qquad P_{l+1} := A_{l+1}\big(\Sigma^{l+1}_{T_l}\big)^{-1}, \tag{51}$$

the spread of a unit mass at the settled spot by one step of the code-book. To restart the front chart we need this density to be strictly positive.

Lemma 5.7 (The new front)**.** *Let $P = A_j(\Sigma^j)^{-1}$ with $\Sigma^j \in (0,\infty)^N$. If k is an interior node, then $(Pe_k)_i > 0$ for every node i and $K^\top P e_k = K^k$. If $k \in \{0, N+1\}$, then $Pe_k = e_k$.*

Proof. Order the nodes with the interior first. The outer columns of A_j are e_0 and e_{N+1} (Corollary 5.1), so

$$A_j = \begin{pmatrix} A_{II} & 0 \\ A_{BI} & I \end{pmatrix}, \qquad P = \begin{pmatrix} A_{II}^{-1} & 0 \\ -A_{BI}A_{II}^{-1} & I \end{pmatrix}.$$

By (15), A_{II} is tridiagonal with strictly negative off-diagonal entries, hence an irreducible non-singular M-matrix, so $A_{II}^{-1} > 0$ entrywise [2]. The block $-A_{BI} \ge 0$ has a positive entry in the row of K^0 (column K^1) and in the row of K^{N+1} (column K^N), so the outer entries of an interior column of P are positive as well. The identity $K^\top P = K^\top$ follows from $K^\top A_j = K^\top$ (Lemma 2.4), and the outer columns are Corollary 5.1. □

Corollary 5.8 (Existence on the whole ladder)**.** *Let b be bounded and locally Lipschitz. On each interval $[T_{l-1}, T_l]$, $l = 1, \dots, M$, let the current front density q^l follow the chart (27) with the settling volatility (31), where $\tau = T_l - t$ and $p \in [\frac{1}{2}, 1)$. Then the model has a solution on $[0, T_M]$, unique given these choices, such that*

(i) every gridded price is a continuous martingale up to its expiry;

(ii) at every T_l the spot sits on a node K^{k_l} and every contract expiring at T_l settles at its payoff;

(iii) if k_l is interior, the model continues on $[T_l, T_{l+1}]$ from the finite state $z = \log q^{l+1}_{T_l}$ and $\lambda^{l+2}, \dots, \lambda^M$ at T_l; if k_l is an outer node, the spot is absorbed there: $q^j_t = e_{k_l}$ for all $j > l$ and $t \ge T_l$.

Proof. By induction on l. On $[0, T_1]$ this is Proposition 4.2 with Theorem 5.6, whose part (ii) also gives the limit λ_{T_1}, finite. Suppose the solution exists on $[0, T_l]$ with λ_{T_l} finite and $q^l_{T_l} = e_k$.

If k is interior, (51) and Lemma 5.7 give a strictly positive new front, so $z = \log q^{l+1}_{T_l}$ is finite, and $K^\top q^{l+1}_{T_l} = K^k = S_{T_l}$, so the spot is continuous. On $[T_l, T_{l+1}]$ the ladder is that of Sections 3–5 with front q^{l+1} and expiries $T_{l+1} < \cdots < T_M$, started from the $\mathcal{F}_{T_l}$-measurable state $(z, \lambda^{l+2}, \ldots, \lambda^M)$. Theorem 5.6 applies with this $\mathcal{F}_{T_l}$-measurable initial condition, the solution of [16, Theorem V.38] depending continuously on its starting point, and with Proposition 4.2 and part (ii) of that theorem gives a unique strong solution on $[T_l, T_{l+1}]$ that settles at T_{l+1} with $\lambda_{T_{l+1}}$ finite. The densities q^j, $j > l+1$, are given by the same recursion on both sides of T_l with continuous inputs, so they are continuous at T_l.

If k is an outer node, Lemma 5.7 gives $q^j_{T_l} = e_k$ for every $j > l$. The constant process $q^j_t \equiv e_k$ satisfies every recursion, since $A_j e_k = e_k$, and is a martingale, so all prices are constant and settle at their payoffs; we freeze λ at λ_{T_l}, on which no price then depends.

Uniqueness on each interval gives uniqueness on $[0, T_M]$. □

6 Finite-step simulation and numerical evidence

We implement the continuous-time construction on the daily SANOS grid used for the empirical study. Every experiment uses 25 daily expiries and 200 interior strikes. The front density is evolved by its exact Gaussian information bridge, DLV squared is evolved in log coordinates, and all later densities are reconstructed from the SANOS systems. The risk-neutral drift is always the drift in (5); no empirical drift is substituted under $\mathbb{Q}$.

The Gaussian bridge gives the exact conditional law of the front density and its settlement node over a finite time interval. In contrast, an ordinary time discretisation of the stochastic DLV equation need not preserve its martingale condition exactly over a full day. We therefore compare one and eight Brownian steps per day, and distinguish algebraic invariants from statistical martingale diagnostics.

6.1 Data, factor rank, and measure change

The fitted sample contains S&P 500 index-option surfaces from 2022 through 29 August 2025. Dynamic parameters are estimated from 710 adjacent market-day pairs whose origin date is no later than 31 December 2024. Within each pair, DLV-squared changes are aligned at fixed cash strike and fixed expiry. Across pairs, the relative-strike grid floats with the contemporaneous spot and option surface.

We use the full 25-by-200 DLV representation. On the held-out 2025 sample its residual root mean square is 0.256, compared with 0.423 for the affine-x 200 representation. Before estimating the covariance, fixed-contract residuals are smoothed by a Gaussian filter with widths one expiry and two strike nodes. The first factor is oriented to the observed front-spot score and the remaining factors are principal components of the orthogonal residual.

Figure 1 shows the resulting factor power. Eight factors explain 72.88% of the smoothed innovation variance and 16 factors explain 84.40%. The first rank above 90% is 26, at 90.25%. Since the experiment was capped at 16 factors, the two expanded models use rank 16 rather than claiming to reach 90%.

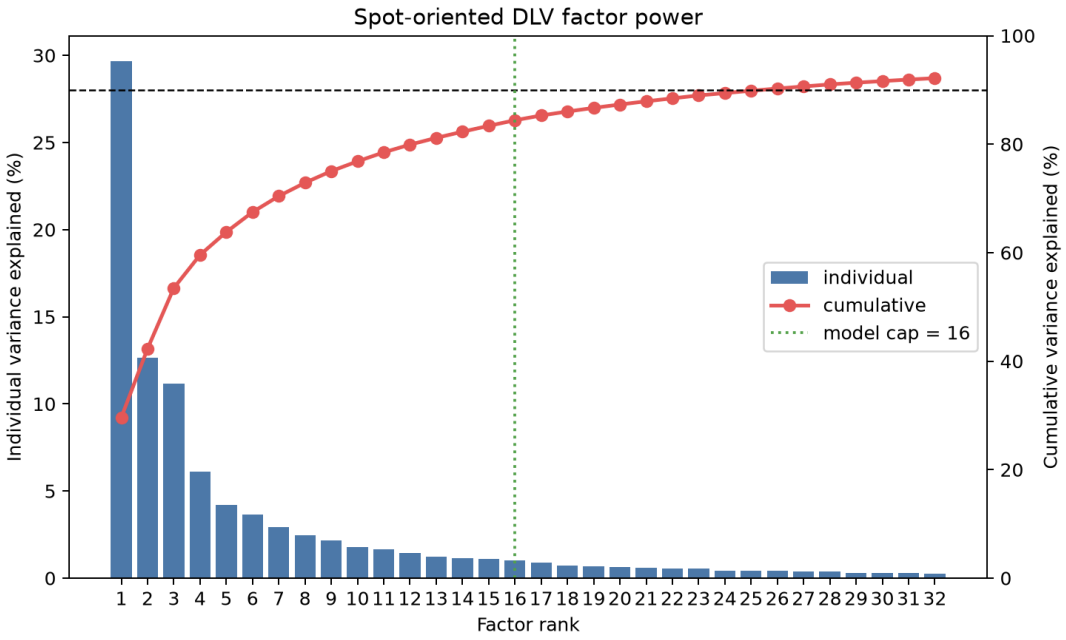


Figure 1: Individual and cumulative variance explained by the spot-oriented DLV factor system. The dotted vertical line is the rank-16 model cap; 26 factors are required to cross the horizontal 90% line.

Figure 2 shows the baseline risk-neutral implied-volatility surface used for the factor illustrations. It is computed from the final fitted state on the same 25-expiry, 200-strike grid used by the simulation.

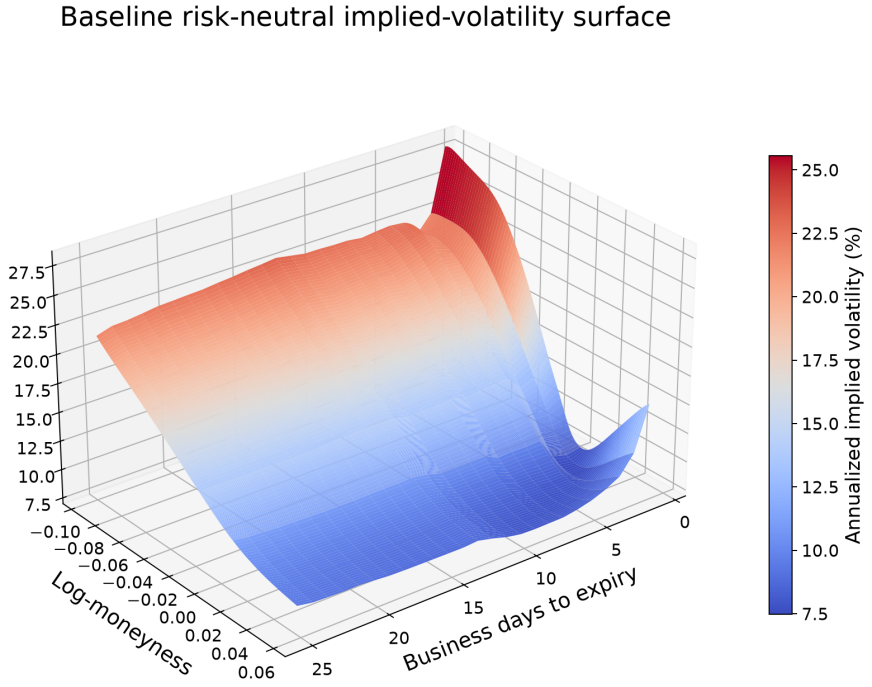


Figure 2: Baseline annualized implied-volatility surface in log-moneyness and business-days-to-expiry coordinates.

Let F_t^1 denote the fitted one-day distribution, including the independent Black factor with annual volatility 5%. The historical spot score is

$$U_t = \Phi^{-1}\big(F_t^1(S_{t+1}/S_t)\big)\,. \tag{52}$$

Its fitted mean and standard deviation are 0.0290 and 1.0026. The first DLV factor is oriented to this score, so the same Brownian coordinate drives the front-density bridge, spot, and the correlated component of the surface shock.

For every historical state, the risk-neutral DLV drift is recomputed from (5). Under the convention $\mathrm{d}W^{\mathbb{Q}} = \mathrm{d}W^{\mathbb{P}} + \lambda\,\mathrm{d}t$, the attainable physical drift is

$$\mu^{\mathbb{P}} = \mu^{\mathbb{Q}} + \xi\lambda. \tag{53}$$

At each historical state t, the state-dependent market price of risk is estimated by

$$\lambda_t^\star = \arg\min_{\lambda\in\mathbb{R}^d} \sum_{j,i} w_t^{j,i} \left[\left(\mu_t^{\mathrm{emp},j,i} - \mu_t^{\mathbb{Q},j,i}\right) - \xi^{j,i}\lambda\right]^2. \tag{54}$$

The weights $w_t^{j,i}$ are proportional to the interior density within each expiry and are normalised so that every expiry has equal total importance. Thus economically negligible tail coordinates do not dominate the fit. The attainable fraction is

$$R^2_{\mathrm{att}} = 1 - \frac{\displaystyle\sum_{t,j,i} w_t^{j,i} \left[\left(\mu_t^{\mathrm{emp},j,i} - \mu_t^{\mathbb{Q},j,i}\right) - \xi^{j,i}\lambda_t^\star\right]^2}{\displaystyle\sum_{t,j,i} w_t^{j,i} \left(\mu_t^{\mathrm{emp},j,i} - \mu_t^{\mathbb{Q},j,i}\right)^2}. \tag{55}$$

It is the fraction of the density-weighted squared empirical $\mathbb{P}$–$\mathbb{Q}$ drift difference lying in the span of the fitted DLV volatility factors. A single constant λ is used for the physical path illustrations below. The independent 5% Black factor is assigned zero risk premium and therefore has the same law under $\mathbb{P}$ and $\mathbb{Q}$.

Calibration quantity	Rank 8	Rank 16
Smoothed variance explained	72.88%	84.40%
Statewise attainable $\mathbb{P}$–$\mathbb{Q}$ drift energy	62.20%	71.95%
Constant-λ attainable drift energy	42.15%	47.24%

Table 1: Factor coverage and attainable physical drift for the two fitted diffusion ranks.

6.2 Brownian simulation and displayed path quantities

Let $m \in \{1, 8\}$ be the number of DLV steps per business day and set $\delta = \Delta t/m$. For every day, substep, and factor we draw independent standard normal vectors $\varepsilon_{n,s} \in \mathbb{R}^d$ and use the genuine Brownian increments

$$\Delta W_{n,s} = \sqrt{\delta}\,\varepsilon_{n,s}, \qquad \sum_{s=1}^{m} \Delta W_{n,s} \sim N(0, \Delta t I_d)\,. \tag{56}$$

At each substep, the state-dependent risk-neutral drift is recomputed and DLV squared is advanced in log coordinates,

$$\Delta \log \Sigma = \left(\mu^{\mathbb{Q}} - \tfrac{1}{2}\|\xi\|^2\right)\delta + \xi \Delta W_{n,s}. \tag{57}$$

The front Gaussian bridge is advanced with the first component of the same Brownian increment. The independent Black factor Y uses one daily Gaussian increment because its constant-volatility GBM transition is exact over the full day. At the daily endpoint, the expired front node is sampled directly from the density at the beginning of that day using the aggregate first-factor increment. This has the same settlement law as recursively revealing the bridge while avoiding accumulation of round-off error. Upper-tail Gaussian interval masses are evaluated as differences of survival probabilities.

The ordinary frozen log-Euler drift is retained whenever the relative-drift increment $x = \mu^{\mathbb{Q}}\delta$ satisfies $|x| \le 0.5$. An unquoted tail node in a development run reached density of order 10^{-24}, at which freezing the relative drift produced $x \simeq 2490$ and overflow. Only for such exceptional increments we use the first-order-consistent safeguard

$$x \longmapsto \operatorname{sgn}(x)\,(0.5 + \log(1 + |x| - 0.5))\,. \tag{58}$$

The stochastic Brownian increment is unchanged. After each update, all later densities are reconstructed with (12). We do not clip DLV squared or replace the SANOS recursion by an additive density update.

Write $Z := XY$ for the tradable underlying, where X is the discrete SANOS factor and Y is the independent Black factor. The path figures below use four matched sets of Brownian draws under $\mathbb{P}$ and $\mathbb{Q}$. They display spot, ATM one- and five-day implied volatility, and the corresponding finite- difference skew

$$\text{Skew}_\tau = \frac{\sigma_{\text{imp}}(K = Z_t e^{h}, \tau) - \sigma_{\text{imp}}(K = Z_t e^{-h}, \tau)}{2h}, \qquad h = 0.01. \tag{59}$$

The path figures illustrate the fitted measure change; four paths are not a statistical validation of the physical model.

For the factor figures, a one-standard-deviation shock means the diffusion-only log-DLV displacement $\pm\xi_f\sqrt{\Delta t}$. Each panel reports the central implied-volatility response

$$\frac{1}{2}\left(\sigma_{\text{imp}}(\log\Sigma+\xi_f\sqrt{\Delta t})-\sigma_{\text{imp}}(\log\Sigma-\xi_f\sqrt{\Delta t})\right). \tag{60}$$

These are also the risk-neutral shock directions because an equivalent measure change alters drift, not diffusion.

6.3 What the risk-neutral audit tests

Each audit starts from the fitted surface on 29 August 2025 and simulates 100,000 independent paths for 25 daily steps on CPU in 64-bit arithmetic. It uses all 200 strikes and five claim families.

(i) For every day, surviving expiry, and node, the daily density return is $q_{n+1}^{j,i}-q_n^{j,i}$. For the settling front density, $q_{n+1}^{j,i}$ is the realised node indicator. These coordinates span all payoffs of the finite X state.

(ii) For every day, surviving expiry, and strike, the daily X-call return is $C_{n+1}^{X,j}(K)-C_n^{X,j}(K)$.

(iii) For every initial expiry and node, the expiry-density return is $\mathbf{1}_{\{X_{T_j}=x_i\}}-q_0^{j,i}$.

(iv) For every initial expiry and strike, the Z-call-to-expiry return is $(Z_{T_j}-K)^+-C_0^{Z,j}(K)$. This test includes the independent Black factor Y.

(v) For every simulated day, the spot return is $Z_{n+1}-Z_n$.

For a zero-cost return R_k, the reported coordinate statistic is

$$z_k=\frac{\overline{R}_k}{\widehat{\text{se}}(\overline{R}_k)}. \tag{61}$$

A coordinate enters the Gaussian screen only if it has at least 20 expected effective observations.[4] Front-density variances use their known conditional Bernoulli variance, front-option variances use their known conditional payoff variance, and to-expiry Z-call variances use analytic Black-mixture second moments. The denominator in (61) is the larger of the predictable and empirical standard errors. This avoids spuriously large Gaussian scores from a single rare payoff. The five-standard-error threshold is a conservative diagnostic screen, not a finite-sample joint test.

For a single pre-specified scalar discounted return R, let $R^{(1)},\dots,R^{(P)}$ be its independent simulated realizations. Its unconditional expected return can be tested directly against any specified value r_0 using

$$\widehat{m}_R=\frac{1}{P}\sum_{p=1}^{P}R^{(p)},\qquad \widehat{\text{se}}(\widehat{m}_R)=\left[\frac{1}{P(P-1)}\sum_{p=1}^{P}\left(R^{(p)}-\widehat{m}_R\right)^2\right]^{1/2},\qquad z_R=\frac{\widehat{m}_R-r_0}{\widehat{\text{se}}(\widehat{m}_R)}. \tag{62}$$

[4] Here "effective observations" means expected informative outcomes under the initial model, not an adjustment for serial dependence: the simulated paths are independent. For a node indicator with probability p and P paths, both the expected number of hits Pp and misses $P(1-p)$ must be at least 20. For a call, the expected exercise count $P\,\mathbb{Q}(\text{payoff}>0)$ must be at least 20.

For a zero-cost risk-neutral return, $r_0 = 0$. This is a test of the unconditional mean only; the variance is a nuisance quantity used to estimate the standard error, not a second null hypothesis. If an analytic or predictable variance is available, the audit can use it conservatively in the same manner as for the coordinate tests. A single pre-specified R also has no multiple-coordinate selection issue, although a rare-event-dominated R still requires enough informative realizations for the Gaussian approximation.

All paths share the same initial market state. Hence the expiry tests are Monte Carlo tests conditional on that initial state. The later daily tests check the unconditional zero means implied by the tower property along the simulated law. They do not estimate $\mathbb{E}[R_{n+1} \mid \mathcal{F}_n]$ separately at every high-dimensional simulated intermediate state. Passing them is necessary evidence for martingality, not a proof of the complete conditional statement.

The audit results are reported by discretisation in Tables 2, 3, 4, and 5. Orange entries mark slight departures from the Gaussian benchmark; red entries mark clear violations of the risk-neutral no-arbitrage screen. Model D, with rank 16 and eight Brownian substeps, passes all five screens. The other models have the violations quantified in their tables and the text following them.

6.4 Model A: rank 8 and one Brownian step per day

The first model uses the original rank-eight diffusion, which explains 72.88% of smoothed DLV innovation variance, and one full-day Brownian increment. The eight risk-neutral one-standard-deviation factor shocks are shown in Figure 3. Figure 4 shows the matched physical and risk-neutral path quantities.

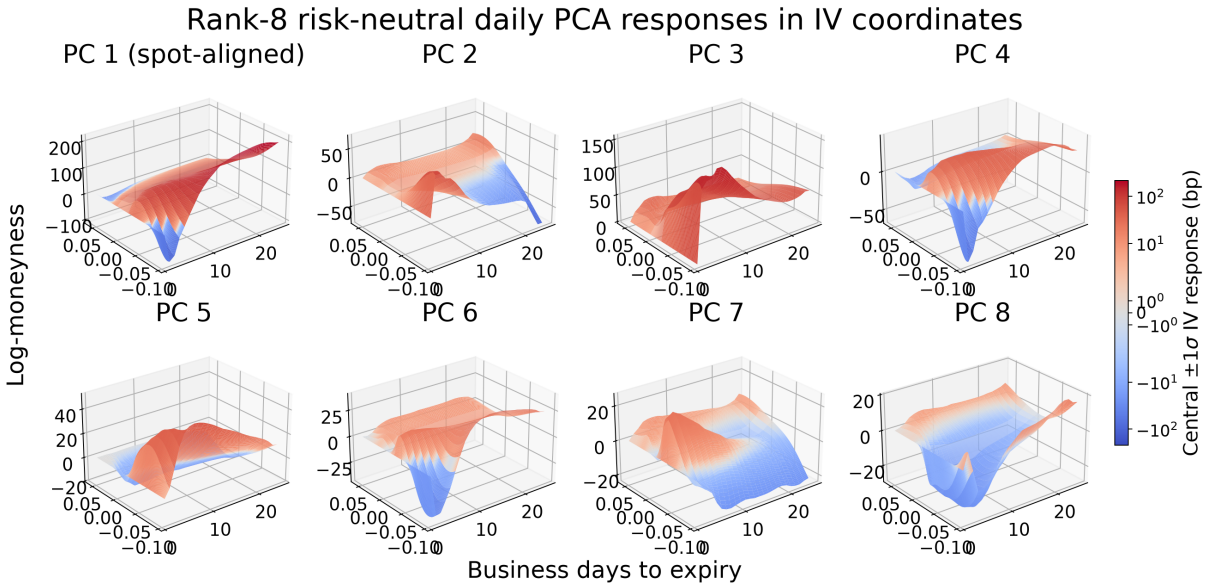


Figure 3: Three-dimensional view of the rank-eight central $\pm 1\sigma$ daily implied-volatility responses. Each panel uses its own vertical scale. The common symmetric-log colour scale preserves the response sign and magnitude while making the lower-power components visible. PC 1 is spot-aligned; the remaining components are orthogonal residual directions.

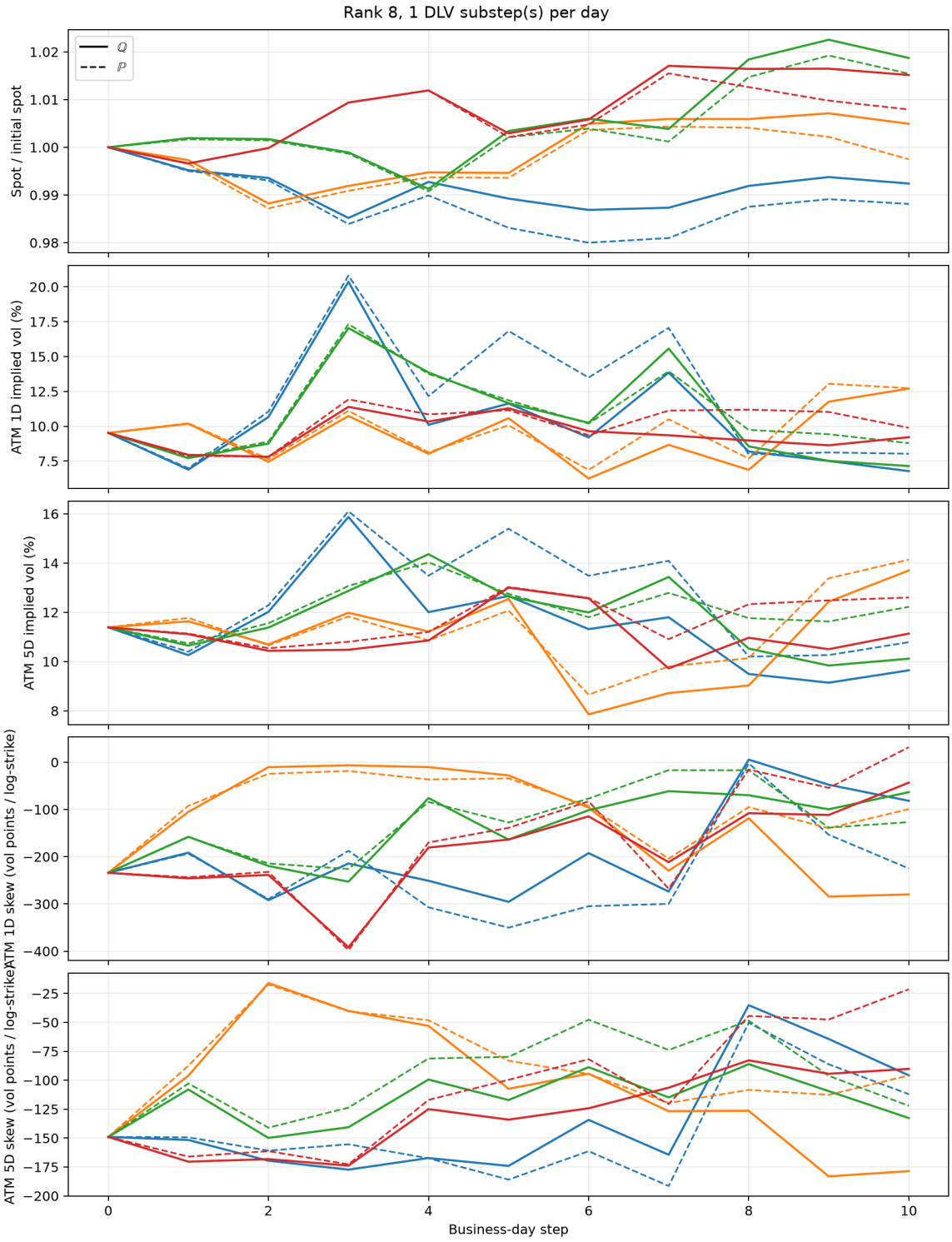


Figure 4: Four coupled $\mathbb{P}/\mathbb{Q}$ path illustrations for rank 8 with one Brownian step per day. Solid and dashed lines of the same colour use the same Gaussian draws under $\mathbb{Q}$ and $\mathbb{P}$, respectively.

The run had no non-finite values; maximum mass and conditional-forward errors were

Claim family	Total	Tested	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$\lvert z\rvert > 5$
Daily X density	65,650	63,723	14.272	3.891	263
Daily X calls	65,000	63,810	5.741	2.246	4
X density to expiry	5,050	4,673	4.220	2.591	0
Z calls to expiry	5,000	4,879	1.991	1.558	0
Daily Z spot	25	25	2.035	1.982	0

Table 2: Rank-eight, one-step, 100,000-path risk-neutral audit. A mean-zero normal would have $99\%|z| = 2.576$.

2.28×10^{-13} and 2.26×10^{-13}. The expiry-density, Z-call, and daily-spot tests pass the screen, but 263 daily density and four daily X-call coordinates exceed five standard errors. The mean corresponding to the largest daily-density statistic is 1.15×10^{-5}. Thus one full-day DLV step is not adequate for the daily martingale conditions, even though its terminal checks appear acceptable. The CPU runtime was 9 minutes 55 seconds.

6.5 Model B: rank 16 and one Brownian step per day

The second model increases the diffusion rank to 16, explaining 84.40% of the smoothed innovation variance, but retains one Brownian increment per day. The additional factors primarily add short-expiry level, skew, and curvature directions. Their shocks are shown in Figure 5, and the coupled paths in Figure 6.

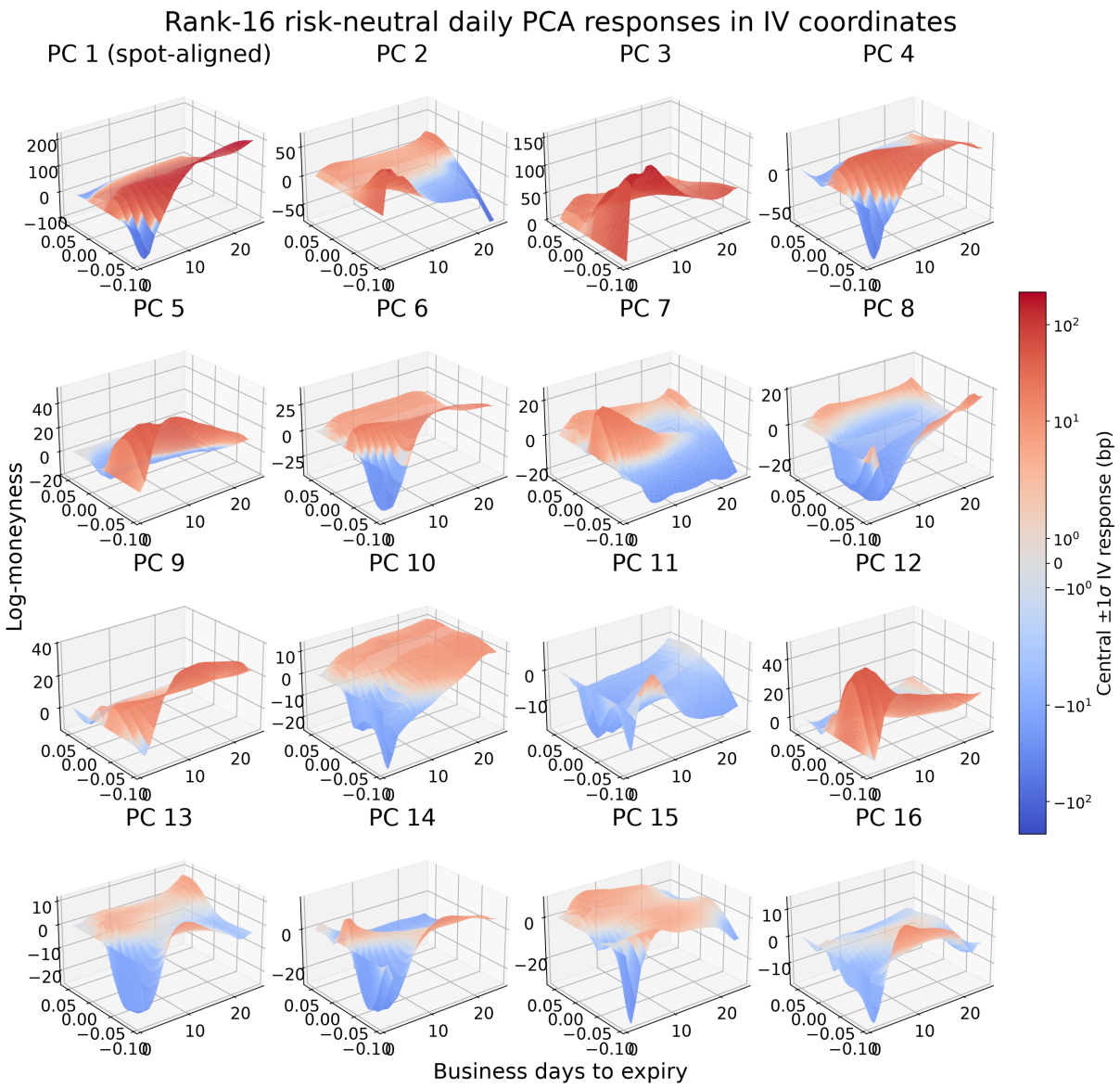


Figure 5: Three-dimensional view of the rank-16 central $\pm 1\sigma$ daily implied-volatility responses. Each panel uses its own vertical scale, while the colour scale is common and symmetric-log. These diffusion shocks are common to $\mathbb{P}$ and $\mathbb{Q}$; the measures differ through drift.

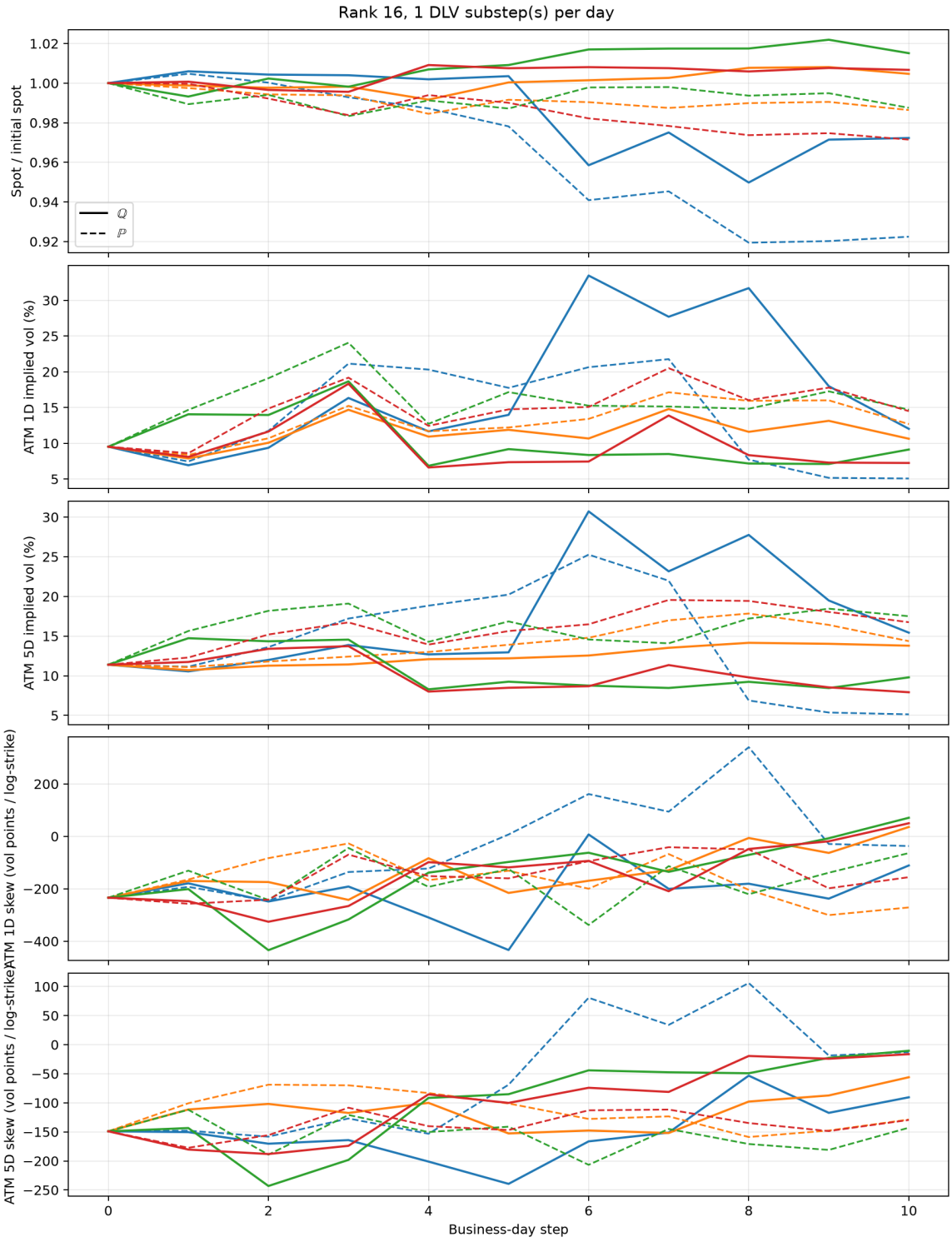


Figure 6: Four coupled $\mathbb{P}/\mathbb{Q}$ path illustrations for rank 16 with one Brownian step per day.

The run had no non-finite values; maximum mass and conditional-forward errors were 2.24×10^{-13} and 2.22×10^{-13}. Nevertheless, 159 daily density and five daily X-call coordinates

Claim family	Total	Tested	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$\lvert z\rvert > 5$
Daily X density	65,650	63,723	15.630	4.000	159
Daily X calls	65,000	63,810	7.083	2.778	5
X density to expiry	5,050	4,673	3.770	2.649	0
Z calls to expiry	5,000	4,879	3.480	3.084	0
Daily Z spot	25	25	1.329	1.316	0

Table 3: Rank-16, one-step, 100,000-path risk-neutral audit. A mean-zero normal would have $99\%|z| = 2.576$.

exceed five standard errors. The largest corresponding means are only 1.32×10^{-5} and -1.67×10^{-6}, but the bias is statistically visible. The CPU runtime was 12 minutes 26 seconds.

6.6 Model C: rank 8 and eight Brownian steps per day

The third model keeps the rank-eight diffusion of Model A but resolves each day with eight independent Brownian subincrements. The factor shocks are unchanged from Figure 3; the coupled paths are shown in Figure 7.

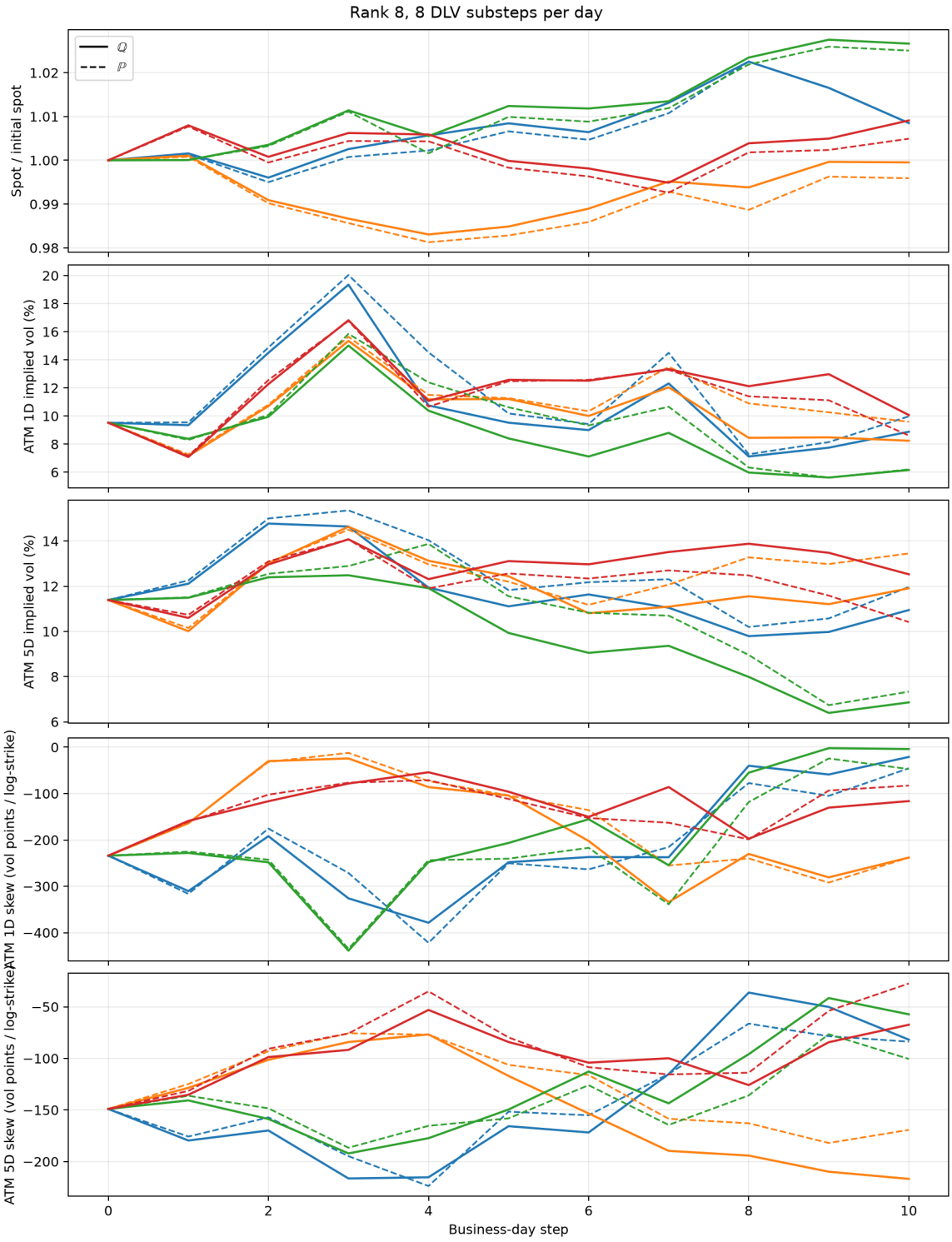


Figure 7: Four coupled $\mathbb{P}/\mathbb{Q}$ path illustrations for rank 8 with eight Brownian substeps per day.

There were no non-finite values; maximum mass and conditional-forward errors were 2.36×10^{-13} and 2.34×10^{-13}. No density, expiry-call, or spot coordinate exceeds five standard errors.

Claim family	Total	Tested	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$\lvert z\rvert > 5$
Daily X density	65,650	63,723	3.814	2.803	0
Daily X calls	65,000	63,810	5.655	2.412	1
X density to expiry	5,050	4,673	3.348	2.465	0
Z calls to expiry	5,000	4,879	2.622	1.698	0
Daily Z spot	25	25	2.255	2.224	0

Table 4: Rank-eight, eight-substep, 100,000-path risk-neutral audit. A mean-zero normal would have $99\%|z| = 2.576$.

One daily X-call coordinate does, with $z = -5.655$ and mean -1.32×10^{-6} in units of initial spot. The CPU runtime was 58 minutes 41 seconds.

6.7 Model D: rank 16 and eight Brownian steps per day

The fourth model combines 16 factors with eight independent Brownian subincrements per day. Its coupled paths are shown in Figure 8.

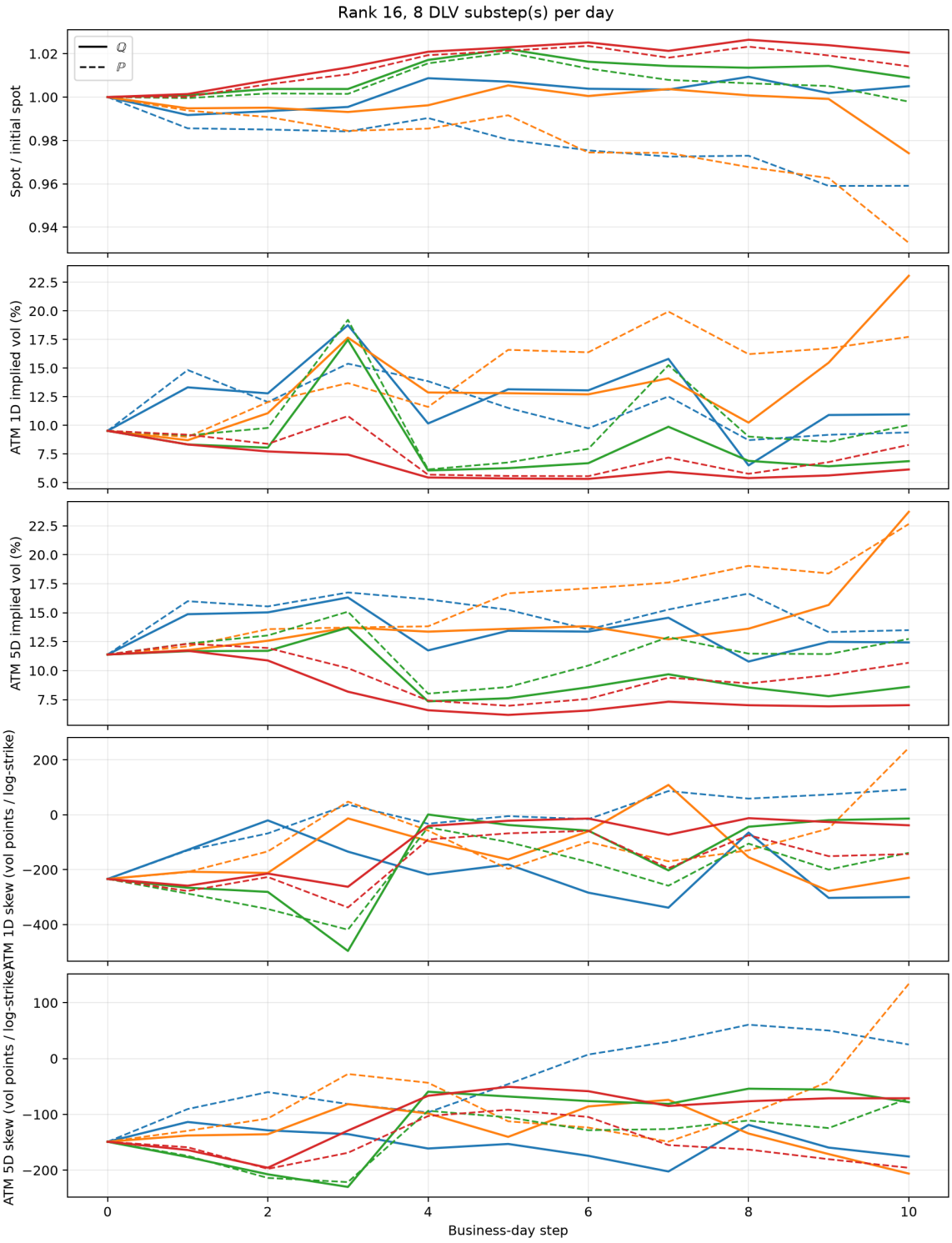


Figure 8: Four coupled $\mathbb{P}/\mathbb{Q}$ path illustrations for rank 16 with eight Brownian substeps per day.

No coordinate exceeds five standard errors. There were no non-finite values; maximum mass and conditional-forward errors were 2.40×10^{-13} and 2.38×10^{-13}. The mean corresponding to

Claim family	Total	Tested	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$\lvert z\rvert > 5$
Daily X density	65,650	63,723	4.068	2.765	0
Daily X calls	65,000	63,810	4.349	2.786	0
X density to expiry	5,050	4,673	3.935	2.428	0
Z calls to expiry	5,000	4,879	2.626	2.483	0
Daily Z spot	25	25	1.741	1.706	0

Table 5: Rank-16, eight-substep, 100,000-path risk-neutral audit. A mean-zero normal would have $99\%|z| = 2.576$.

the largest daily X-call statistic was -8.01×10^{-7} in units of initial spot. The CPU runtime was 1 hour 3 minutes 30 seconds.

6.8 Interpretation

Models A and B show that one daily Brownian increment is not sufficient for the nonlinear DLV/SANOS discretisation: increasing factor rank from 8 to 16 does not remove the daily density bias. Eight substeps remove every density violation: Model C retains one daily-call exceedance, whereas Model D passes every screen. The paired comparisons A–C and B–D identify intraday time resolution as the dominant numerical correction, with the higher factor rank removing the final detectable call remainder in this experiment.

This remains numerical evidence for the continuous-time construction, not a claim of exact finite-step martingality. Exact martingality for a specified discrete shock law would require solving the nonlinear branch correction in (13). Nor do the physical path figures constitute an out-of-sample validation of $\mathbb{P}$; they visualize the attainable drift and its difference from $\mathbb{Q}$. The implementation retains the stated continuous-time dynamics, evolves positive DLV squared in log coordinates, and reports explicitly where a coarse finite-step approximation fails.

7 What is open

1. *An off-lattice terminal spot.* By Remark 4.3 the chart represents every $s \in [K^0, K^{N+1}]$ at expiry, and the state that does so is $\varpi(s)$, carried by the two nodes straddling s. Whether a front volatility exists driving q^1 to $\varpi(s)$ for s strictly inside a cell, rather than to a single node as (31) does, is open. The obstruction is that exact settlement forces $q^1_{T_1} = \varpi(S_{T_1})$ and hence $\int_0^{T_1} (\eta^{1,i})^2 \mathrm{d}t < \infty$ for both surviving nodes, which by the argument of Proposition 4.2 they cannot both satisfy when the volatility is proportional to distance from the spot. Until it is answered the model of Proposition 4.2 pins, as [21] does, and his question of a front that does not pin the spot remains open.

2. *The continuum limit.* The constant $(M-1)(N+1)$ grows with the grid. Whether $\sup_N \Theta_j < \infty$ on a class of realistic surfaces, which is what a continuum statement would need, is open.

8 Implementation Commets

Data and estimation. The implementation uses the daily 25-expiry by 200-strike data set. The historical DLV-squared innovations are aligned locally at fixed cash strike and fixed expiry

over each one-day pair. Across different day pairs the surface and its support are allowed to float with the market. There are 710 training pairs through 31 December 2024. We use the full DLV state rather than the affine-200 state: the respective 2025 out-of-sample residual root mean squares are 0.256 and 0.423. Gaussian smoothing with widths one expiry and two strike nodes is applied before a rank-eight functional principal-component fit; the retained factors explain 72.88% of the smoothed innovation variance.

The historical spot shock is calculated exactly as proposed above,

$$Z_t = \Phi^{-1}\big(F_t^1(S_{t+1}/S_t)\big)\,,$$

where F_t^1 is the fitted one-day front distribution including the independent 5% Black smoothing process. Its fitted mean and standard deviation are 0.0290 and 1.0026. The first DLV factor is oriented to this score. Under $\mathbb{Q}$ the DLV drift is recomputed from (5) at every numerical substep. Under $\mathbb{P}$ we impose

$$\mu^{\mathbb{P}} = \mu^{\mathbb{Q}} + \xi\lambda$$

and project the empirical conditional DLV drift onto this attainable subspace. The statewise projection explains 62.20% of the weighted drift energy; a single constant λ used for simulation explains 42.15%. The independent 5% Black process is assigned zero risk premium and is therefore identical under $\mathbb{P}$ and $\mathbb{Q}$.

Numerical update. The front density is advanced with the exact Gaussian information bridge and the same first Brownian factor that drives spot leverage. Gaussian interval probabilities in the upper tail are evaluated as differences of survival probabilities, avoiding subtraction of two CDF values close to one. At the end of a day the expired front is sampled directly from the density at the start of that day using the aggregate first-factor increment. This is algebraically equivalent to recursively revealing the bridge, while avoiding accumulation of floating-point error in the settlement law.

DLV squared is evolved in log coordinates with eight uniform intraday substeps. The drift and front-density volatility are recomputed at each substep and the future density tower is reconstructed with (12). The ordinary frozen log-Euler drift increment $x = \mu\delta$ is used whenever $|x| \le 0.5$. One simulation path exposed a density of order 10^{-24} for which freezing the relative drift produced $x \simeq 2490$ and numerical overflow. Only for such exceptional increments we use the tail safeguard

$$x \longmapsto \operatorname{sgn}(x)\,(0.5 + \log(1 + |x| - 0.5))\,.$$

It leaves ordinary steps unchanged and has the same infinitesimal limit as the stated SDE. We do not clip DLV squared or alter the SANOS density recursion.

Several alternatives were rejected during implementation. An additive Euler update of the densities can leave the positive convex-order cone, and a simple antithetic recentering has the same problem. A balanced explicit–implicit DLV drift multiplier removed the overflow but introduced measurable bias on ordinary paths. A grid refined in squared remaining time did not improve a matched experiment. These failures are numerical rather than changes to the continuous-time drift identity.

CPU validation. A coupled 512-path, 25-day $\mathbb{P}/\mathbb{Q}$ pilot produced no non-finite values. Mass and forward errors were below 1.3×10^{-13}. The final risk-neutral audit used 100,000 paths,

eight substeps and all 200 strikes. It tested 65,650 daily density returns, 65,000 daily latent-option returns, 5,050 density-to-expiry returns, 5,000 option-to-expiry returns including the Black smoothing process, and all 25 daily spot returns. The run had no non-finite values; its maximum mass and forward errors were 2.61×10^{-13} and 2.58×10^{-13}.

For rare claims, a Gaussian statistic based only on predictable variance can be misleading. In one reproduced example the exercise count was 96 against 96.44 expected, but one exceptionally large payoff gave a predictable- variance score of 9.18 while the realised studentised score was 0.98. Accordingly, the audit retains the analytically known conditional variance but uses the larger of predictable and empirical standard errors for inference. All expiry-density, expiry-option, daily-density and daily-spot statistics were below five standard errors, with maxima 3.53, 2.83, 3.83 and 2.27.

Two of the 63,810 statistically identifiable daily latent-option coordinates remained above five standard errors. Their worst statistic was 6.21, but the corresponding mean return was only -5.27×10^{-8}. Thus the implementation is stable and the residual bias is economically tiny, but this experiment does not claim exact martingality for an arbitrary finite time step. This is consistent with the discussion following (12): the continuous-time drift condition removes the infinitesimal drift, not the nonlinear finite-step bias. Exact martingality under a specified discrete shock law would require solving the branchwise nonlinear correction (13).

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